

The number of normal measures, revisited

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Abstract

We present a new version of the Friedman–Magidor theorem: For every measurable cardinal κ and cardinal $\tau \leq \kappa^{++}$, there exists a forcing extension $V \subseteq V[G]$ such that any normal measure $U \in V$ on κ has exactly τ distinct lifts in $V[G]$, and every normal measure on κ in $V[G]$ arises as such a lift. This version differs from the original Friedman–Magidor theorem in several notable ways. First, the new technique does not involve forcing over canonical inner models or rely on any fine-structural tools or assumptions, allowing it to be applied in the realm of large cardinals beyond the current reach of the inner model program. Second, in the case where $\tau \leq \kappa^+$, all lifts of a normal measure $U \in V$ on κ to $V[G]$ have the same ultrapower; as an application, a nontrivial model of the Weak Ultrapower Axiom is constructed. Finally, the technique generalizes to a version of the Friedman–Magidor theorem for extenders. An additional advantage is that the forcing used is notably simple, relying only on nonstationary support product forcing.

1 Introduction

The analysis of measures in forcing extensions plays a central role in the theory of large cardinals. A major question in this area was whether the existence of a model of ZFC with exactly two normal measures follows from the consistency of ZFC with a measurable cardinal. This question was answered affirmatively by the following landmark theorem of Friedman and Magidor.

Theorem 1.1. *(Friedman–Magidor theorem, [11]) Suppose $V = L[U]$ is the minimal inner model in which there is a measurable cardinal κ , and let $\tau \leq \kappa^{++}$ be a cardinal. Then in a cardinal preserving forcing extension, κ is measurable and carries exactly τ normal measures.*

In their work, Friedman and Magidor identified the various components that contribute to the construction of normal measures in forcing extensions. Their forcing notion includes, for each such component, a specialized feature designed to control and constrain it. This naturally led to the orchestration of several

forcing techniques, each playing an essential role in the proof. Indeed, Theorem 1.1 stands out for the range of methods it brings together, including forcing over canonical inner models, nonstationary support iterations, generalized Sacks forcing, and interleaved self-coding posets. These techniques have since enabled a variety of further applications, including [5], [4], [6], [1], [2], and [7].

The central result of this paper is a significantly simpler and more transparent proof of the Friedman–Magidor theorem. Our approach dispenses entirely with inner model theory, self-coding, and generalized Sacks forcing. Instead, it relies solely on a product forcing with nonstationary support.

Similarly to the original Friedman–Magidor proof, we concentrate on the case where the expected number τ of normal measures is at most κ^+ . Achieving $\tau = \kappa^{++}$ is done by performing the Kunen–Paris forcing (see [21]).

The main theorem of this paper is the following:

Theorem 1.2. *Suppose that κ is a measurable cardinal and $\tau \leq \kappa^+$ is a cardinal. There exists a poset $\mathbb{R}^\tau$ which preserves κ, κ^+ and the measurability of κ , such that, whenever $G \subseteq \mathbb{R}^\tau$ is generic over V , the following properties hold:*

1. *Every normal measure $U \in V$ on κ has exactly τ distinct lifts $\langle U_\eta^* : \eta < \tau \rangle$ to normal measures on κ in $V[G]$, all of which have the same ultrapower over $V[G]$.*
2. *Every normal measure $W \in V[G]$ on κ has the form U_η^* for some normal measure $U \in V$ and $\eta < \tau$.*

Also, assuming GCH in V , $\mathbb{R}^\tau$ preserves cardinals.

In the case where $\tau < \kappa$, $\mathbb{R}^\tau$ has the form $\text{Add}(\omega, 1) * \mathbb{P}^\tau$, where $\mathbb{P}^\tau$ consists of partial functions $f : \kappa \rightarrow \tau$ whose support is nowhere stationary, ordered by inclusion.¹ In the case where $\tau \in \{\kappa, \kappa^+\}$, $\mathbb{P}^\tau$ is slightly modified using canonical functions (see Definition 3.2 below).

As mentioned above, one of the main advantages of Theorem 1.2 is that it does not involve forcing over canonical inner models. In the original Friedman–Magidor proof, forcing over $L[U]$ played a significant role in two ways. First, by assuming that $L[U]$ is the core model and forcing over it, every normal measure on κ in the generic extension restricts to an iterated ultrapower of $L[U]$. (This is the starting point of a sequence of classical results in inner model theory, culminating in a well-known theorem of Schindler, see [24].) Second, canonical inner models possess fine structure, which Friedman and Magidor utilized by incorporating self-coding posets into their forcing construction. (See [11] for further details on self-coding and its role in constraining the number of normal measures.)

While these techniques demonstrate the utility of inner model theory and fine structure in forcing theory, forcing over canonical inner models can be restrictive—particularly because it does not apply at levels of the large cardinal

¹The purpose of the initial Cohen forcing is to ensure that $\mathbb{R}^\tau$ has a gap below κ in Hamkins’ sense of [17] (see Theorem 3.7).

hierarchy not yet reached by the inner model program. Since our proof of Theorem 1.2 avoids the use of inner model theory, it can be employed to obtain consistency results at the level of strongly compact and supercompact cardinals, such as the following:

Theorem 1.3 (Apter-Kaplan-Poveda, see [3]). *Assume the consistency of a supercompact cardinal and the linearity of the Mitchell order.*

1. *It is consistent that there exists a supercompact cardinal, and there are exactly two normal measures on the least measurable cardinal.*
2. *Assume in addition the existence of a measurable cardinal above the supercompact cardinal. Then it is consistent that there is a supercompact cardinal, and the least measurable cardinal above it carries exactly two normal measures.*

Our forcing notions $\mathbb{P}^\tau$ (for $\tau < \kappa$) have several additional applications. In Section 4, we classify all possible lifts of U^n to the generic extension obtained by forcing with $\mathbb{P}^\tau$, where $U \in V$ is a normal measure on κ and $n < \omega$.² When this analysis is carried out in the case where $V = L[U]$, it allows for a complete description of the σ -complete ultrafilters in the extension. This yields a ZFC model with an interesting measure structure:

Theorem 1.4. *Assume that $V = L[U]$, and let κ be the unique measurable cardinal. Fix a cardinal $\tau < \kappa$. Let $G \subseteq \mathbb{P}^\tau$ be generic over V . Then in $L[U][G]$:*

1. *There are τ -many normal measures on κ , $\langle U_\eta^* : \eta < \tau \rangle$, and all of them have the same ultrapower.*
2. *For every σ -complete ultrafilter $W \in L[U][G]$, the ultrapower embedding $j_W^{L[U][G]} : L[U][G] \rightarrow \text{Ult}(L[U][G], W)$ is a finite iterated ultrapower of $L[U][G]$ by normal measures.*
3. *Every ultrapower of $L[U][G]$ (via a σ -complete ultrafilter) is isomorphic to $\text{Ult}(L[U][G], (U_0^*)^n)$ for some $n < \omega$.*

Theorem 1.4 is related to the Weak Ultrapower Axiom. Recall that Goldberg's Ultrapower Axiom (UA) states that for any pair of σ -complete ultrafilters U, W , there are σ -complete ultrafilters $U^* \in M_W$ and $W^* \in M_U$,³ such that $M_{U^*}^{M_W} = M_{W^*}^{M_U}$ and $j_{W^*} \circ j_U = j_{U^*} \circ j_W$. The Weak Ultrapower Axiom is obtained from UA by removing the demand that $j_{W^*} \circ j_U = j_{U^*} \circ j_W$. One of the consequences of Theorem 1.4 (for $1 < \tau < \kappa$) is that a model of the Weak UA and the negation of UA⁴ can be constructed, assuming the consistency of a measurable cardinal.

²For the definition of the measure U^n , see Definition 4.1.

³Here and below, for a σ -complete ultrafilter U , we denote by M_U the transitive collapse of $\text{Ult}(V, U)$, and by $j_U : V \rightarrow M_U$ the corresponding ultrapower embedding.

⁴The negation of UA holds in the model of Theorem 1.4 since, by a theorem of Goldberg, UA implies the linearity of the Mitchell order. See [14] for more details.

Theorem 1.5. *Weak UA + $\neg$ UA is equiconsistent with a single measurable cardinal.*

Theorem 1.5 slightly improves a previous result by Goldberg and Benhamou (see [9]), who constructed a nontrivial model of Weak UA assuming the consistency of a measurable limit of measurable cardinals.

Finally, building on the techniques introduced earlier, we generalize the Friedman–Magidor framework from normal measures to extenders in Theorem 5.4. A typical application of this theorem is illustrated by the following example:

Theorem 1.6. *Assume GCH, and let κ be a $(\kappa + 2)$ -strong cardinal. Then for every cardinal $\tau \leq \kappa^{++}$, there is a cardinal preserving forcing extension $V[G]$ in which every (κ, κ^{++}) -extender $E \in V$ witnessing that κ is $(\kappa + 2)$ -strong lifts to exactly τ pairwise nonequivalent⁵ (κ, κ^{++}) -extenders in $V[G]$. In $V[G]$, all such lifts of E have the same ultrapower and continue to witness the $(\kappa + 2)$ -strongness of κ . Moreover, every (κ, κ^{++}) -extender witnessing that κ is $(\kappa + 2)$ -strong in $V[G]$ arises as one of these lifts.*

Goldberg observed that the bound κ^{++} on the number of (κ, κ^{++}) -extenders having the same ultrapower is optimal under GCH (see Theorem 5.14). Similar results can be obtained for extenders with higher degrees of strongness.

The structure of this paper is as follows: In Section 2, we cover the basic properties of nonstationary support products. In Section 3, we prove Theorem 1.2. In Section 4, we discuss lifting iterates of U^n for a normal measure U and $n < \omega$, and prove Theorems 1.4 and 1.5. In Section 5, we state and prove a Friedman–Magidor theorem for extenders (Theorem 5.4), and apply it to prove Theorem 1.6. In Section 6, we discuss several open questions.

Throughout the paper, we use the following convention in forcing: If a condition p extends a condition q , we write $q \leq p$, and the weakest condition in a forcing notion $\mathbb{P}$ is denoted by $0_{\mathbb{P}}$. We will also occasionally make standard abuse of notation by omitting the “check” sign from “check names” in statements in the forcing language. In addition, we will, from time to time, make standard abuse of notation by omitting “dots” over forcing terms. We will, however, on occasion, also include both checks and dots in our forcing iterations. The rest of the notation is standard.

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⁵Two extenders are *equivalent* if they induce the same ultrapower embedding.

extension have the same ultrapower. Ben-Neria suggested the key idea that led to the uniqueness of normal ultrapowers, and also to the current simplified version of the forcing in Definition 3.2 below. The author is deeply grateful to both of them for their support and encouragement throughout the preparation of this work.

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2 Nonstationary support products

In this section, we briefly develop the basic framework of product forcing with a nonstationary support. All the ideas previously appeared in one form or other in [11]. Our presentation is influenced by [8] and [1].

Definition 2.1. A set of ordinals A is *nowhere stationary*⁶ if for every inaccessible cardinal α , $A \cap \alpha$ is nonstationary in α .

Definition 2.2. (Products with nonstationary support) Fix a cardinal κ , and let $I \subseteq \kappa$ be an unbounded set of ordinals. For every $\alpha \in I$, let $\mathbb{Q}_\alpha$ be a forcing notion. The *nonstationary support product* $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ consists of conditions p which are functions with domain I , such that for every $\alpha \in I$, $p(\alpha) \in \mathbb{Q}_\alpha$, and the set

$$\text{supp}(p) = \{\alpha \in I : p(\alpha) \neq 0_{\mathbb{Q}_\alpha}\}$$

is nowhere stationary.

The ordering is defined by saying that p extends q if for every $\alpha \in \text{supp}(q)$, $p(\alpha)$ extends $q(\alpha)$ in $\mathbb{Q}_\alpha$ (in particular, $\text{supp}(q) \subseteq \text{supp}(p)$).

Assuming that $\beta < \kappa$ and $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ is a nonstationary support product, $\mathbb{P}$ can be naturally factored to the form $\mathbb{P} \restriction \beta \times \mathbb{P} \restriction \beta$, where $\mathbb{P} \restriction \beta = \prod_{\alpha \in \beta \cap I}^{NS} \mathbb{Q}_\alpha$ and $\mathbb{P} \restriction \beta = \prod_{\alpha \in I \setminus \beta}^{NS} \mathbb{Q}_\alpha$, by identifying a condition $p \in \mathbb{P}$ with the pair $(p \restriction (I \cap \beta), p \restriction (I \setminus \beta))$. We denote in this case $p \restriction \beta = p \restriction (I \cap \beta)$ and $p \restriction \beta = p \restriction (I \setminus \beta)$. To streamline the notation, we will often denote the poset $\mathbb{P} \restriction \beta$ by $\mathbb{P}_\beta$.

The most important feature of nonstationary support products or iterations is their fusion property.

Lemma 2.3 (The Fusion Lemma). *Let κ be a limit of inaccessible cardinals, and let $I \subseteq \kappa$ be an unbounded set of inaccessible cardinals below κ . Let $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ be a nonstationary support product. Assume that:*

⁶We will adopt this terminology, although the phrase *nonstationary in inaccessible cardinals* would be more accurate in this context. For our purposes, this distinction is irrelevant, since the nowhere stationary sets we consider will consist solely of inaccessible cardinals. For such sets, being nonstationary in inaccessible cardinals implies being nonstationary in every ordinal of uncountable cofinality (see [7, Lemma 3.24]).

(1) For every $\alpha \in I$, $\text{rank}(\mathbb{Q}_\alpha) < \min(I \setminus (\alpha + 1))$.

(2) For every $\alpha \in I$, $\mathbb{Q}_\alpha$ is α -closed.

Then $\mathbb{P}$ satisfies the **Fusion Property**, that is, given:

- a condition $p \in \mathbb{P}$,
- a sequence $\langle d(\alpha) : \alpha < \kappa \rangle$ of dense open subsets of $\mathbb{P}$,

there exist $p^* \geq p$ and a club $C \subseteq \kappa$ (if $\text{cof}(\kappa) = \omega$, C is an unbounded cofinal ω -sequence) such that, for every $\alpha \in C$, the set

$$\{r \in \mathbb{P} \upharpoonright (\alpha + 1) : r \frown p^* \setminus (\alpha + 1) \in d(\alpha)\}$$

is dense and open in $\mathbb{P} \upharpoonright (\alpha + 1)$ above $p^* \upharpoonright (\alpha + 1)$.

Proof. We first take care of the case where κ is singular. Fix an inaccessible cardinal $\nu \in (\text{cof}(\kappa), \kappa)$, and a continuous, cofinal, increasing sequence $\langle \nu_i : i < \text{cof}(\kappa) \rangle$ in κ , such that $\nu_0 > \nu$. We argue that there exists $s \geq p \setminus (\nu + 1)$ such that for every $i < \text{cof}(\kappa)$ and $r \in \mathbb{P} \upharpoonright (\nu + 1)$, there exists $r' \geq r$ such that $r' \frown s \in d(\nu_i)$. Indeed, to construct such s , fix an enumeration $\langle \langle i_j, r_j \rangle : j < \zeta \rangle$ of $\text{cof}(\kappa) \times \mathbb{P}_{\nu+1}$ (where $\zeta = |\mathbb{P}_{\nu+1}|$), and construct an increasing sequence $\langle s_j : j \leq \zeta \rangle \subseteq \mathbb{P} \setminus (\nu + 1)$, such that:

- $s_0 = p \setminus (\nu + 1)$.
- For every $j < \zeta$, $s_{j+1} \geq s_j$ is chosen such that there exists $r' \geq r_j$ such that $r' \frown s_{j+1} \in d(\nu_{i_j})$.
- For every limit $j \leq \zeta$, s_j is chosen as an upper bound of $\langle s_{j'} : j' < j \rangle$. More formally, s_j is constructed such that $\text{supp}(s_j) = \bigcup \{\text{supp}(s_{i'}) : i' < j\}$, and for every $\alpha \in \text{supp}(s_j)$, $s_j(\alpha)$ is an upper bound of $\{s_{i'}(\alpha) : i' < j, \alpha \in \text{supp}(s_{i'})\}$. Note that $\text{supp}(s_j)$ is nowhere stationary, since the club filter on every inaccessible cardinal above ν is ζ^+ -complete (hence, for every inaccessible cardinal $\mu > \nu$, $\text{supp}(s_j) \cap \mu$ is nonstationary in μ).

Finally, let $s = s_\zeta$. Then s has the property that for every $i < \text{cof}(\kappa)$, the set

$$\{r \in \mathbb{P}_{\nu+1} : r \frown s \in d(\nu_i)\}$$

is dense open. Let $p^* = (p \upharpoonright (\nu + 1)) \frown s$. Then $p \leq p^* \in \mathbb{P}$, and for every $i < \text{cof}(\kappa)$, the set

$$\{r \in \mathbb{P} \upharpoonright (\nu_i + 1) : r \frown p^* \setminus (\nu_i + 1) \in d(\nu_i)\}$$

is dense and open in $\mathbb{P} \upharpoonright (\nu_i + 1)$ above $p^* \upharpoonright (\nu_i + 1)$. Thus, p^* and $C = \{\nu_i : i < \text{cof}(\kappa)\}$ are as desired.

Therefore, assume from now on that κ is regular. We construct by induction sequences $\langle p_i : i < \kappa \rangle \subseteq \mathbb{P}$, $\langle \nu_i : i < \kappa \rangle \subseteq \kappa$ such that $\langle p_i : i < \kappa \rangle$ is a fusion sequence with respect to $\langle \nu_i : i < \kappa \rangle$, in the sense that the following properties hold:

1. $\langle \nu_i : i < \kappa \rangle$ is a continuous, cofinal sequence in κ .
2. $\langle p_i : i < \kappa \rangle$ is an increasing sequence of conditions extending p , such that, for every $i \leq j < \kappa$,
 - (a) $p_j \upharpoonright (\nu_i + 1) = p_i \upharpoonright (\nu_i + 1)$.
 - (b) $\nu_j \notin \text{supp}(p_i)$.
3. For every $i < \kappa$,

$$\{r \in \mathbb{P} \upharpoonright (\nu_i + 1) : r \restriction (p_{i+1} \setminus (\nu_i + 1)) \in d(\nu_i)\}$$

is a dense open subset of $\mathbb{P} \upharpoonright (\nu_i + 1)$.

Assuming that such sequences have been constructed, we can set $p^* = \bigcup_{i < \kappa} p_i \upharpoonright (\nu_i + 1)$. Since $C = \{\nu_i : i < \kappa\}$ is a club disjoint from $\text{supp}(p^*)$, p^* is a legitimate condition in $\mathbb{P}$.⁷ Then p^* and C are as desired: Indeed, for every $i < \kappa$, the fact that the set

$$\{r \in \mathbb{P} \upharpoonright (\nu_i + 1) : r \restriction p_{i+1} \setminus (\nu_i + 1) \in d(\nu_i)\}$$

is dense open, and the fact that $p^* \restriction (\nu_i + 1) \geq p_{i+1} \restriction (\nu_i + 1)$, imply that the set

$$\{r \in \mathbb{P} \upharpoonright (\nu_i + 1) : r \restriction p^* \setminus (\nu_i + 1) \in d(\nu_i)\}$$

is dense open.

Thus, it suffices to prove that sequences $\langle p_i : i < \kappa \rangle$ and $\langle \nu_i : i < \kappa \rangle$ with the desired properties can be constructed. We define these sequences inductively, along with an auxiliary sequence $\langle C_i : i < \kappa \rangle$ of clubs in κ , such that, in addition to the above requirements:

4. For every $i < \kappa$, C_i is disjoint from $\text{supp}(p_i)$.
5. For every $i < j < \kappa$, $\nu_j \in C_i$ and $C_j \subseteq C_i$.

We proceed to the construction. For $i = 0$, take $p_0 = p$, $\nu_0 = 0$, and C_0 any club disjoint from $\text{supp}(p_0)$.

For the successor step, assume that p_i, ν_i, C_i have been constructed. First set $\nu_{i+1} = \min(C_i \setminus (\nu_i + 1))$. Define $p_{i+1} \geq p_i$ such that:

- $p_{i+1} \upharpoonright (\nu_{i+1} + 1) = p_i \upharpoonright (\nu_{i+1} + 1)$ (in particular, $\nu_{i+1} \notin \text{supp}(p_{i+1})$).
- $p_{i+1} \restriction (\nu_{i+1} + 1)$ extends $p_i \restriction (\nu_{i+1} + 1)$, and satisfies that

$$\{r \in \mathbb{P} \upharpoonright (\nu_{i+1} + 1) : r \restriction (p_{i+1} \setminus (\nu_{i+1} + 1)) \in d(\nu_{i+1})\}$$

is a dense open subset of $\mathbb{P} \upharpoonright (\nu_{i+1} + 1)$

⁷Note that we require that $\text{supp}(p^*)$ is nonstationary in any inaccessible cardinal $\lambda < \kappa$ as well. This follows immediately, since given such λ , $p^* \restriction \lambda = p_i \restriction \lambda$ for any large enough $i < \kappa$, and p_i is already known to have a nowhere stationary support.

The second bullet is achieved similarly to the parallel argument in the singular case: Starting from an enumeration $\langle r_j : j < \zeta \rangle$ of $\mathbb{P}_{\nu_{i+1}+1}$, construct $p_{i+1} \setminus (\nu_{i+1}+1)$ as an upper bound of an increasing sequence of conditions $\langle s_j : j < \zeta \rangle$, such that $r_j \cap s_{j+1} \in d(\nu_{i+1})$ for every $j < \zeta$. Note that conditions (1), (2) of the hypotheses of Lemma 2.3 ensure that the forcing $\mathbb{P} \setminus (\nu_{i+1}+1)$ is sufficiently closed, allowing the construction of such a condition p_{i+1} . Finally, let $C_{i+1} \subseteq C_i$ be a club in κ disjoint from $\text{supp}(p_{i+1})$.

For the limit step, assume that $i < \kappa$ and $\langle p_j, \nu_j, C_j : j < i \rangle$ were constructed. First, set $\nu_i = \sup\{\nu_j : j < i\}$. Let $p_i \in \mathbb{P}$ be a condition such that:

- $p_i \restriction \nu_i = \bigcup_{j < i} p_j \restriction (\nu_j + 1)$.
- $\nu_i \notin \text{supp}(p_i)$.
- $p_i \setminus (\nu_i + 1)$ extends all the conditions $\langle p_j \setminus (\nu_i + 1) : j < i \rangle$, and satisfies that

$$\{r \in \mathbb{P} \restriction (\nu_i + 1) : r \cap (p_i \setminus (\nu_i + 1)) \in d(\nu_i)\}$$

is a dense open subset of $\mathbb{P} \restriction (\nu_i + 1)$.

Note that, as above, $\mathbb{P} \setminus (\nu_i + 1)$ is closed enough to satisfy the third requirement. Also, by our construction, ν_i is a limit point of each club C_j for $j < i$, and thus $\nu_i \notin \text{supp}(p_j)$ for every $j < i$. This ensures that p_i extends p_j for $j < i$. Finally, let $C_i \subseteq \bigcap_{j < i} C_j$ be a club disjoint from $\text{supp}(p_i)$. This concludes the inductive construction, which concludes the proof. $\square$

Corollary 2.4. *Let $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ be as in Lemma 2.3. Suppose that $p \in \mathbb{P}$, and let $\dot{f}$ be a $\mathbb{P}$ -name forced by p to be a function from κ to the ordinals. Then there exist $q \geq p$, a club $C \subseteq \kappa$, and a function $F \in V$ such that $C \subseteq \text{dom}(F)$, for every $\alpha \in C$, $|F(\alpha)| \leq |\mathbb{P}_{\alpha+1}|$ (in particular, $|F(\alpha)| < \min(I \setminus (\alpha + 1))$), and $q \Vdash \dot{f}(\alpha) \in F(\alpha)$.*

Proof. Apply Lemma 2.3 to the sequence $\langle d(\alpha) : \alpha < \kappa \rangle$, where each $d(\alpha)$ is the dense open set of conditions deciding the value of $\dot{f}(\alpha)$. Let $q \geq p$ be the condition obtained and $C \subseteq \kappa$ the club obtained, and define, for every $\alpha \in C$,

$$F(\alpha) = \{\gamma \in \text{ON} : \exists r \geq q \restriction (\alpha + 1) \left((r \cap (q \setminus (\alpha + 1))) \Vdash \dot{f}(\alpha) = \check{\gamma} \right)\}.$$

$\square$

The assumptions of Lemma 2.3 suffice to ensure that κ is preserved after forcing with $\mathbb{P}$. But we can obtain even more.

Corollary 2.5. *Let $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ be as in Lemma 2.3. Then there is no cofinal function $f : \kappa \rightarrow \kappa^+$ in $V[G]$.*

Proof. Let $p \in \mathbb{P}$ and $\dot{f}$ be a $\mathbb{P}$ -name such that p forces that $\dot{f}$ is an increasing function from κ to κ^+ . We wish to prove that $\text{Im}(\dot{f})$ is forced by an extension of p to be bounded in κ^+ . By Corollary 2.4, there exist $q \geq p$ and a club $C \subseteq \kappa$

such that for every $\alpha \in C$, $q \Vdash \dot{f}(\alpha) \in F(\alpha)$ and $|F(\alpha)| < \min(I \setminus (\alpha + 1)) < \kappa$. Let $\gamma = \sup\{\sup(F(\alpha)) : \alpha \in C\} + 1$. Then $\gamma < \kappa^+$, and since C is unbounded in κ and $p \Vdash \text{"}\dot{f} \text{ is increasing"}$, $q \Vdash \sup(\text{Im}(\dot{f})) < \gamma$. $\square$

In particular, Corollary 2.5 implies that $\mathbb{P}$ preserves κ^+ . Assuming further that GCH holds in V , and that each component forcing $\mathbb{Q}_\alpha$ preserves cardinals, we can strengthen the conclusion to full preservation of cardinals by $\mathbb{P}$. Without the GCH assumption, however, this need not hold (see Claim 6.5).

Corollary 2.6. *Assume GCH. Let $\mathbb{P} = \prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$ be as in Lemma 2.3. Assume that for every $\alpha \in I$, $\Vdash_{\mathbb{P}_\alpha} \text{"}\mathbb{Q}_\alpha \text{ preserves cardinals"}$. Then $\mathbb{P}$ preserves cardinals.*

Proof. Assume otherwise, and let μ be the least cardinal which is not preserved. It follows that μ is a successor cardinal. We can't have $\mu \geq \kappa^{++}$, since $|\mathbb{P}| = \kappa^+$ by our assumption that $2^\kappa = \kappa^+$. So $\mu = \lambda^+$ for some $\lambda \leq \kappa$, and by Corollary 2.5, we can assume that $\lambda < \kappa$. The forcing $\mathbb{P} \restriction (\lambda + 1)$ is more than λ^+ -closed, and if $\lambda \in I$, the forcing $\mathbb{Q}_\lambda$ preserves λ^+ . So if $\mu = \lambda^+$ is collapsed, this is due to $\mathbb{P} \restriction \lambda$. If λ is not a limit point of I , then by GCH, $|\mathbb{P} \restriction \lambda| \leq \lambda$ and thus μ is preserved. So we can assume that λ is a limit point of I , and $\mathbb{P}_\lambda$ collapses $\mu = \lambda^+$. This contradicts Corollary 2.5. $\square$

We conclude this section with the following classical application of Corollary 2.4.

Corollary 2.7. *Assume that κ is regular. For every club $C \subseteq \kappa$ in $V[G]$, there exists a club $D \subseteq \kappa$ in V such that $D \subseteq C$.*

Proof. Let $f \in V[G]$ be the increasing enumeration of C . Apply Corollary 2.4 to find, in V , a function $g : \kappa \rightarrow \kappa$ such that g dominates f on a (ground model) club. By that, we mean that there exists a club $E \subseteq \kappa$ in V such that, for every $i \in C$, $f(i) < g(i)$. Since f is increasing and κ is regular, we can assume that g is increasing and dominates f everywhere (namely $E = \kappa$). Now define D to be the club of closure points of g . Then $D \subseteq C$, since every $\alpha \in D$ is a limit point of $C \cap \alpha$: Indeed, $C \cap \alpha$ is unbounded in α , since for every $i < \alpha$, $f(i) \leq g(i) < \alpha$ and $i \leq f(i) \in C$. $\square$

3 The splitting forcing

The goal of this section is to prove Theorem 1.2. Throughout this section, suppose that κ is a measurable cardinal in V . Denote by I the set of inaccessible cardinals below κ . Fix a cardinal $\tau \leq \kappa^+$.

Let $\langle f_\eta : \eta < \kappa^+ \rangle \subseteq \kappa^\kappa$ be a sequence of functions such that f_η is the η -th canonical function. Recall that the functions $\langle f_\eta : \eta < \kappa^+ \rangle$ can be defined by induction on $\eta < \kappa^+$, as follows: Let $f_0(\alpha) = 0$ for all $\alpha < \kappa$, and for every $\eta < \kappa$, let $f_{\eta+1}(\alpha) = f_\eta(\alpha) + 1$. If η is a limit ordinal, let $\langle \eta_i : i < \text{cof}(\eta) \rangle$ be an increasing sequence cofinal in η . Define $f_\eta(\alpha) = \sup_{i < \text{cof}(\eta)} f_{\eta_i}(\alpha)$ in the case $\text{cof}(\eta) < \kappa$, and $f_\eta(\alpha) = \sup_{i < \alpha} f_{\eta_i}(\alpha)$ in the case $\text{cof}(\eta) = \kappa$ (see [18] for more background on canonical functions).

We also denote by f_{κ^+} the cardinal successor function, namely $f_{\kappa^+}(\alpha) = \alpha^+$ for all $\alpha < \kappa$.

We will make repeated use of the following property of the functions $\langle f_\eta : \eta \leq \kappa^+ \rangle$ throughout this section.

Claim 3.1. *Suppose that $V \subseteq V[G]$ is a forcing extension which preserves α, α^+ for every inaccessible cardinal $\alpha \leq \kappa$, as well as the measurability of κ . Let $W \in V[G]$ be a normal measure on κ . Then for every $\eta \leq \kappa^+$, $j_W(f_\eta)(\kappa) = \eta$.*

Proof. An easy induction, which we leave to the interested reader, shows that for every $\eta < \kappa^+$, $j_W(f_\eta)(\kappa) = \eta$. It remains to consider the case $\eta = \kappa^+$. Since $(\alpha^+)^{V[G]} = (\alpha^+)^V$ for all inaccessible cardinals $\alpha < \kappa$, the function f_{κ^+} agrees with the cardinal successor function computed in $V[G]$ on the set I of inaccessible cardinals. As $I \in W$, it follows that

$$j_W(f_{\kappa^+})(\kappa) = (\kappa^+)^{\text{Ult}(V[G], W)} = (\kappa^+)^{V[G]} = (\kappa^+)^V,$$

as desired. $\square$

Let us define the main forcing notion in this paper. We refer to it as “the splitting forcing”.

Definition 3.2 (The splitting forcing). Define

$$\mathbb{P}^\tau = \{f \in \prod_{\alpha \in X} f_\tau(\alpha) : X = \text{dom}(f) \text{ is a nowhere stationary subset of } I\}$$

ordered by inclusion, that is, f extends g if and only if $g \subseteq f$.

The forcing notion $\mathbb{P}^\tau$ can be viewed as a nonstationary support product $\prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$, where for every $\alpha < \kappa$, $\mathbb{Q}_\alpha$ is trivial unless α is inaccessible. For α inaccessible, $\mathbb{Q}_\alpha = \{0_{\mathbb{Q}_\alpha}\} \cup f_\tau(\alpha)$ is the atomic forcing notion that satisfies:

- $0_{\mathbb{Q}_\alpha}$ is the weakest condition of $\mathbb{Q}_\alpha$.
- The ordinals in $f_\tau(\alpha)$ form a maximal antichain in $\mathbb{Q}_\alpha$.

In particular, a generic for $\mathbb{Q}_\alpha$ is basically a choice of an ordinal in $f_\tau(\alpha)$.

Remark 3.3. In the case where $\tau < \kappa$, we will always assume that the function f_τ from Claim 3.1 is constant with value τ . In this case, $\mathbb{P}^\tau$ admits the following simpler presentation:

$$\mathbb{P}^\tau = \{f : X \rightarrow \tau : X = \text{dom}(f) \text{ is a nowhere stationary subset of } I\}.$$

Lemma 3.4. *Assume that $G \subseteq \mathbb{P}^\tau$ is generic over V . For every $\eta < \tau$, define*

$$S_\eta = \{\alpha \in I : (\cup G)(\alpha) = f_\eta(\alpha)\}.$$

Then $\vec{S} = \langle S_\eta : \eta < \tau \rangle$ is a sequence of stationary subsets of I . If $\tau < \kappa$, $\vec{S}$ is a partition of I , and $V[G] = V[\vec{S}]$.

Proof. Fix $\eta < \tau$, and let $C \in V[G]$ be a club in κ . By Corollary 2.7, there exists in V a club $D \subseteq \kappa$ such that $D \subseteq C$. Given a condition $p \in \mathbb{P}^\tau$, we can pick $\alpha \in D \setminus \text{supp}(p)$ due to the nonstationary support, and extend p to $p \cup \{(\alpha, f_\eta(\alpha))\}$. Thus, by density, there exist $q \in G$ and $\alpha \in D$ such that $q(\alpha) = f_\eta(\alpha)$. Thus, in $V[G]$, $\alpha \in C \cap S_\eta$.

If $\tau < \kappa$, then each $f_\eta(\alpha)$ simply equals η , and $S_\eta, S_{\eta'}$ for $\eta \neq \eta'$ are disjoint. Clearly, $\bigcup_{\eta < \tau} S_\eta = I$. Hence, $\vec{S}$ is a partition of I . Also, G can be read from $\vec{S}$, since for every condition p , $p \in G$ if and only if

$$\forall \alpha \in \text{dom}(p) \ (p(\alpha) = \eta \iff \alpha \in S_\eta).$$

□

Definition 3.5. Assume that W is an ultrafilter on κ and $X \subseteq W$. We say that X *generates* W if $W = \{A \subseteq \kappa : \exists B \in X (B \subseteq A)\}$.

Next, we show that every normal measure $U \in V$ on κ generates, together with one of the stationary sets S_η , a normal measure on κ in $V[G]$.

Lemma 3.6. *Let $U \in V$ be a normal measure on κ . Let $G \subseteq \mathbb{P}^\tau$ be generic over V . Then for every $\eta < \tau$, $U \cup \{S_\eta\}$ generates a normal measure U_η^* on κ in $V[G]$. Furthermore, the U_η^* -s have the same ultrapower over $V[G]$.*

Proof. Since τ is fixed, we let $\mathbb{P} = \mathbb{P}^\tau$ throughout this proof. Fix $\eta < \tau$. The forcing $j_U(\mathbb{P})$ is, by elementarity, a nonstationary support product $\prod_{\alpha \in j_U(I)}^{NS} \mathbb{Q}_\alpha^{M_U}$, where for each $\alpha \in j_U(I) \cap \kappa = I$, $\mathbb{Q}_\alpha^{M_U} = \mathbb{Q}_\alpha$, and for each $\alpha \in j_U(I) \setminus \kappa$, $\mathbb{Q}_\alpha^{M_U}$ is an atomic forcing that chooses an ordinal below $j_U(f_\tau)(\alpha)$. In particular, $\kappa \in j_U(I)$, and $\mathbb{Q}_\kappa^{M_U}$ is an atomic forcing that chooses an ordinal below τ .

Let

$$H_\eta = \{q \in j_U(\mathbb{P}) : \exists p \in G \ (q \subseteq j_U(p) \cup \{(\kappa, \eta)\})\}.$$

Note that for every $p \in \mathbb{P}$, $\kappa \notin \text{supp}(j_U(p))$. Therefore $j_U(p) \cup \{(\kappa, \eta)\}$ is a condition, and its support is a nowhere stationary subset of $j_U(\kappa)$, so $H_\eta \subseteq j_U(\mathbb{P})$. Furthermore, the elements of H_η are mutually compatible, and clearly $j_U[G] \subseteq H_\eta$. We argue that $H_\eta \subseteq j_U(\mathbb{P})$ is generic over M_U .

Fix $E \subseteq j_U(\mathbb{P})$ dense open in M_U . Let $\langle e(\alpha) : \alpha < \kappa \rangle$ be a sequence of dense open subsets of $\mathbb{P}$ such that $E = [\alpha \mapsto e(\alpha)]_U$. Recall that $\mathbb{P}$ can be viewed as a nonstationary support product. Apply the Fusion Lemma 2.3 to find a condition $q \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$,

$$\{r \in \mathbb{P}_{\alpha+1} : r^\frown (q \setminus (\alpha+1)) \in e(\alpha)\}$$

is dense in $\mathbb{P}_{\alpha+1}$. This means that, in M_U , the set

$$\{r \in j_U(\mathbb{P})_{\kappa+1} : r^\frown (j_U(q) \setminus (\kappa+1)) \in E\}$$

is dense in $j_U(\mathbb{P})_{\kappa+1}$. Since ${}^\kappa M_U \subseteq M_U$, $j_U(\mathbb{P})_\kappa = \mathbb{P}_\kappa$. Therefore, G is generic for $j_U(\mathbb{P})_\kappa$ over M_U . Moreover, the ordinal η dictates a generic for $\mathbb{Q}_\kappa^{M_U}$ over

$M_U[G]$. Thus, there exists $p \in G$ such that $p \smallfrown \langle \langle \kappa, \eta \rangle \rangle \smallfrown (j_U(q) \setminus (\kappa + 1)) \in E$. By extending p if necessary inside G , we can assume that p extends q . Since E is open, we deduce that $j_U(p) \cup \{ \langle \kappa, \eta \rangle \} \in E \cap H_\eta$, as desired.

Since $j_U[G] \subseteq H_\eta$, we can now use Silver's criterion to lift the embedding j_U to an elementary embedding from $V[G]$ to $M_U[H_\eta]$. By standard arguments (see [10]), the lifted embedding is an ultrapower embedding with respect to a normal measure U_η^* . Indeed, U_η^* is the normal measure derived from the lifted embedding using κ as a seed. In other words, for $X \in V[G]$, we have $X \in U_\eta^*$ if and only if there exists a condition in H_η that forces that $\kappa \in j_U(\dot{X})$ (where $\dot{X}$ is a $\mathbb{P}$ -name for X). Equivalently, there exists a condition $p \in G$ such that

$$j_U(p) \cup \{ \langle \kappa, \eta \rangle \} \Vdash \kappa \in j_U(\dot{X}).$$

This implies that $\{ \alpha < \kappa : p \cup \{ \langle \alpha, f_\eta(\alpha) \rangle \} \in U \}$, and

$$\{ \alpha < \kappa : p \cup \{ \langle \alpha, f_\eta(\alpha) \rangle \} \Vdash \check{\alpha} \in \dot{X} \} \cap \{ \alpha < \kappa : (\cup G)(\alpha) = f_\eta(\alpha) \} \subseteq X,$$

so $U \cup \{ S_\eta \}$ generates U_η^* .

Note that for every $\eta, \zeta < \tau$, H_η, H_ζ are mutually definable: One is obtained from the other by changing the generic value at κ . This implies that all the measures U_η^* ($\eta < \tau$) have the same ultrapower. $\square$

We now proceed to prove that every normal measure of $V[G]$ on κ has the form U_η^* for some $\eta < \tau$ and normal measure $U \in V$ on κ . The main tool that assists bypassing inner model theoretic arguments here is Hamkins' Gap Forcing Theorem.

Theorem 3.7 (Hamkins' Gap Forcing Theorem [17]). *Assume that $\mathbb{P}$ is a forcing notion which has a gap at some cardinal δ . That is, $\mathbb{P}$ can be factored in the form $\mathbb{P} = \mathbb{P}_0 * \mathbb{P}_1$, where $\mathbb{P}_0$ is nontrivial, $|\mathbb{P}_0| < \delta$, and $\Vdash_{\mathbb{P}_0}$ " $\mathbb{P}_1$ is $(\delta + 1)$ -strategically closed."*⁸ *Let $G \subseteq \mathbb{P}$ be generic over V , and assume that $j : V[G] \rightarrow M^*$ is an elementary embedding, with critical point $\kappa > \delta$, such that $M^* \subseteq V[G]$ and M^* is closed under δ -sequences of its elements that belong to $V[G]$. Then:*

1. *M^* has the form $M[H]$, where $M \subseteq V$ and $H = j(G)$ is $j(\mathbb{P})$ -generic over M . Furthermore, $M = V \cap M[H]$.*
2. *If j is definable in $V[G]$ from parameters, then $j \restriction V : V \rightarrow M$ is definable in V from parameters.*

Remark 3.8. We remark that M^* having the form $M[H]$ already follows from Laver's Ground Model Definability Theorem (see [22, Theorem 3]). However, the inclusion $M \subseteq V$ is nontrivial and need not hold for forcings without a gap (see, for example, [13, Subsection 5.2]).

⁸See [10, Definition 5.15] for the definition of strategic closure of posets.

Remark 3.9. Nonstationary support iterated forcings usually satisfy the assumptions of the Gap Forcing Theorem. Our splitting forcing $\mathbb{P}^\tau$, however, is a nonstationary support product rather than an iteration. It is unclear whether the conclusions of the Gap Forcing Theorem apply directly to $\mathbb{P}^\tau$. To address this, we modify $\mathbb{P}^\tau$ by considering $\mathbb{R}^\tau = \text{Add}(\omega, 1) * \mathbb{P}^\tau$, which has a gap at $\delta = \omega_1$.

We now have all the ingredients to conclude the proof of Theorem 1.2.

Proof of Theorem 1.2. Let $\mathbb{R}^\tau = \text{Add}(\omega, 1) * \mathbb{P}^\tau$. Let $G = g * G' \subseteq \mathbb{R}^\tau$ be generic over V , where g is $\text{Add}(\omega, 1)$ -generic over V , and G' is $\mathbb{P}^\tau$ -generic over $V[g]$. By the Lévy–Solovay theorem [23], every normal measure on κ in V generates a normal measure on κ in $V[g]$, and conversely, every normal measure on κ in $V[g]$ arises in this way. Given $U \in V$ a normal measure, denote by $U' \in V[g]$ the normal measure it generates. As above, given $\eta < \tau$, let $S_\eta = \{\alpha < \kappa : (\cup G')(\alpha) = f_\eta(\alpha)\}$. By Lemma 3.6, for every $\eta < \tau$, $U' \cup \{S_\eta\}$ generates a measure $U'_\eta \in V[g * G']$, and all the measures $\langle U'_\eta : \eta < \tau \rangle$ have the same ultrapower. We can slightly abuse the notation and write U_η^* instead of U'_η . Each such U_η^* is generated by $U \cup \{S_\eta\}$ in $V[g * G']$, since every set in U' contains a set from U . This concludes the proof of clause 1 of Theorem 1.2, since again by Lemma 3.6, the U_η^* -s all have the same ultrapower over $V[G]$.

We proceed and prove clause 2. Let $W \in V[G]$ be a normal measure on κ , where as before, G is viewed as a generic $g * G' \subseteq \text{Add}(\omega, 1) * \mathbb{P}^\tau$. By Hamkins' Gap Forcing Theorem, $U = W \cap V \in V$, since $U = \{X \subseteq \kappa : \kappa \in (j_W \upharpoonright V)(X)\}$. U is normal because W is normal in $V[G]$. Let $\eta = j_W(\cup G')(\kappa)$. Using again the normality of W and Claim 3.1, $\eta < j_W(f_\tau)(\kappa) = \tau$.⁹ In particular, $S_\eta \in W$. Therefore $U \cup \{S_\eta\} \subseteq W$, and, by the previous paragraph, $W = U_\eta^*$. This concludes the proof of clause 2 of Theorem 1.2.

Finally, if GCH holds in V , then it continues to hold in $V[g]$, and over this model, $\mathbb{P}^\tau$ preserves cardinals by Corollary 2.6. $\square$

An analogous result to Theorem 1.2 holds for nonstationary support iterations in place of products (see [3]). However, it is unclear whether, in this setting, all the lifts $\langle U_\eta^* \mid \eta < \tau \rangle$ yield the same ultrapower.

We finish this section by proving that the splitting forcing $\mathbb{P}^\tau$ preserves the Mitchell order. As a corollary, $\mathbb{R}^\tau$ preserves the Mitchell order as well.

Lemma 3.10. *Assume that $U_0 \triangleleft U_1$ are normal measures on κ in V . Fix $\eta, \zeta < \tau$ and $G \subseteq \mathbb{P}^\tau$ generic over V . Then $(U_0)_\eta^* \triangleleft (U_1)_\zeta^*$.*

Proof. We begin by observing that because ${}^\kappa M_{U_1} \subseteq M_{U_1}$, $\mathbb{P}^\tau \in M_{U_1}$. Therefore, since $U_0 \triangleleft U_1$, we can lift U_0 to $\left((U_0)_\eta^*\right)^{M_{U_1}[G]}$ by forcing with $\mathbb{P}^\tau$ over M_{U_1} .

It suffices to prove that $\left((U_0)_\eta^*\right)^{M_{U_1}[G]} = (U_0)_\eta^*$.

⁹Note that in the case where $\tau = \kappa^+$, we needed here that $\mathbb{R}^\tau$ preserves α^+ for every inaccessible $\alpha \leq \kappa$, in order to ensure that f_τ remains the cardinal successor function in the sense of $V[g * G']$ (see the proof of Claim 3.1). This follows from the fact that $\mathbb{P}^\tau$ preserves α^+ for every $\alpha \in I$, by Corollary 2.5.

Both ultrafilters $(U_0)_\eta^*$, $\left((U_0)_\eta^*\right)^{M_{U_1}[G]}$ are generated by $U_0 \cup \{S_\eta\}$, but the former ultrafilter belongs to $V[G]$, and the latter belongs to $M_{U_1}[G]$. Our goal is to prove that those ultrafilters coincide (this would conclude the proof, since $M_{U_1}[G]$ is a ground of $\text{Ult}(V[G], (U_1)_\zeta^*)$, by the proof of Lemma 3.6). Thus, it remains to verify that $V[G]$ and $M_{U_1}[G]$ have the same subsets of κ . It's clear that $M_{U_1}[G] \subseteq V[G]$, so we just need to argue that $(\mathcal{P}(\kappa))^{V[G]} \subseteq (\mathcal{P}(\kappa))^{M_{U_1}[G]}$.

Let $\dot{X} \in V$ be a $\mathbb{P}^\tau$ -name for a subset of κ . For every $\alpha < \kappa$, let

$$e(\alpha) = \{r \in \mathbb{P}^\tau : r \restriction (\alpha + 1) \Vdash \exists X_\alpha \subseteq \alpha \ (r \restriction (\alpha + 1) \Vdash \dot{X} \cap \alpha = X_\alpha)\}.$$

Each $e(\alpha)$ is a dense open subset of $\mathbb{P}^\tau$. Indeed, given $\alpha < \kappa$ and a condition $r \in \mathbb{P}^\tau$, let $r^* \geq r$ be such that $r^* \restriction (\alpha + 1) = r \restriction (\alpha + 1)$, and in $V^{(\mathbb{P}^\tau)^{\alpha+1}}$, $r^* \restriction (\alpha + 1)$ decides $\dot{X} \cap \alpha$. The latter is possible since $\langle \mathbb{P}^\tau \restriction (\alpha + 1), \leq \rangle$ is more than α -closed. It follows that $r \leq r^* \in e(\alpha)$. This proves that $e(\alpha)$ is dense, and it's straightforward to verify that $e(\alpha)$ is open.

By the Fusion Lemma 2.3, there exist $p \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$,

$$p \restriction (\alpha + 1) \Vdash \exists X_\alpha \subseteq \alpha \ (p \restriction (\alpha + 1) \Vdash \dot{X} \cap \alpha = X_\alpha).$$

For each $\alpha \in C$, let $\dot{X}_\alpha$ be a canonical $\mathbb{P}_{\alpha+1}$ -name for X_α as a subset of α . Note that $\langle \dot{X}_\alpha : \alpha \in C \rangle \in M_{U_1}$, and since ${}^\kappa M_{U_1} \subseteq M_{U_1}$, $p \in M_{U_1}$. Since $p \in G$, we can compute $(\dot{X})_G$ in $M_{U_1}[G]$ from G and $\langle \dot{X}_\alpha : \alpha \in C \rangle$. $\square$

Remark 3.11. There are two main limitations on τ in this section: τ is a cardinal, and $\tau \leq \kappa^+$.

1. Regarding the requirement $\tau \leq \kappa^+$: Recall that our analysis of $\mathbb{P}^\tau$ in this section involved viewing $\mathbb{P}^\tau$ as a nonstationary support product $\prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$, where $\mathbb{Q}_\alpha \in V_\kappa$ for all $\alpha \in I$. However:
 - If $2^\kappa = \kappa^+$, cardinals $\tau \geq \kappa^{++}$ cannot be represented by a function $f_\tau : \kappa \rightarrow \kappa$. Thus, there are no natural alternatives for the posets $\mathbb{Q}_\alpha$ which also belong to V_κ .
 - If $2^\kappa > \kappa^+$, it is unclear under which circumstances $\mathbb{P}^\tau$ preserves cardinals. In fact, forcing with $\mathbb{P}^\tau$ can collapse κ^{++} (see Claim 6.5).

These issues, specifically in the case where $\tau = \kappa^{++}$, motivate Question 6.1.

2. Regarding the requirement that τ is a cardinal: We imposed this assumption, as our focus is on the number of normal measures. Nevertheless, all the results of this section remain valid if one allows τ to be an arbitrary ordinal $\leq \kappa^+$. On the other hand, the situation becomes more subtle when τ is an ordinal above κ^+ . In this case, the function f_τ representing τ is no longer canonical, and it depends on the measure U we wish to lift. Furthermore, the classification of normal measures in the forcing extension is

more complicated than the one given in Theorem 1.2. We will study this more general framework in Section 5, and classify the normal measures in the generic extension in Theorem 5.16.

4 Lifting iterates of a normal measure

Our next goal is to prove Theorem 1.4, which provides a characterization of all possible lifts of U^n for a normal measure U on κ and $n < \omega$. Recall that U^n is defined as follows:

Definition 4.1. Assume that U is a normal measure on a measurable cardinal κ , and fix $1 \leq n < \omega$. Recall that the measure U^n on $[\kappa]^n$ is defined recursively, by setting $U^1 = U$, and for every $1 \leq n < \omega$,

$$U^{n+1} = \{X \subseteq [\kappa]^{n+1} : \{\langle \xi_0, \dots, \xi_{n-1} \rangle \in [\kappa]^n : \{\xi < \kappa : \langle \xi_0, \dots, \xi_{n-1}, \xi \rangle \in X\} \in U\} \in U^n\}.$$

We also extend the above definition for $n = 0$, by setting U^0 to be the principal ultrafilter on the empty set, namely $U^0 = \{\emptyset\}$.

It is a standard fact that the ultrapower $\text{Ult}(V, U^n)$ is identical to the n -th iterated ultrapower of V with respect to U (in the case where $n = 0$, $\text{Ult}(V, U^0) = V$).

The main technical result of this section is a classification of all possible lifts of the measures U^n (for $n < \omega$) in the generic extension (see Corollary 4.5). To the best of our knowledge, no analogous classification exists for the original Friedman-Magidor forcing construction from [11].

We maintain the same notation as in the previous section, including the measurable cardinal κ and the forcing notion $\mathbb{P}^\tau$, which we denote simply by $\mathbb{P}$ throughout this section.¹⁰

In this section, we restrict attention to the case $\tau < \kappa$ only, due to technical difficulties that arise when applying the same proof technique in the cases $\tau \in \{\kappa, \kappa^+\}$.

We denote by κ_i for $i < \omega$ the images of κ under the iterated ultrapowers with U , that is, for each $i < \omega$, $\kappa_i = j_{U^i}(\kappa)$.

Lemma 4.2. *Fix a normal measure U on κ , $n < \omega$, and a sequence $\langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$. Assume that for every $\vec{\nu} = \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n$, $e(\vec{\nu}) = e(\nu_0, \dots, \nu_{n-1}) \subseteq \mathbb{P}$ is a dense open set. Then for every $p \in \mathbb{P}$, there exists $r \geq p$ such that*

$$\{\vec{\nu} \in [\kappa]^n : r \cup \{\langle \nu_0, \eta_0 \rangle, \langle \nu_1, \eta_1 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle\} \in e(\vec{\nu})\} \in U^n.$$
¹¹

Proof. We begin with the case $n = 1$. Let p be a condition. By the Fusion Lemma 2.3, there exists a condition $q \geq p$ and a club $C \subseteq \kappa$ such that for every $\nu \in C$,

$$\{r \in \mathbb{P}_{\nu+1} : r \cup (q \setminus (\nu + 1)) \in e(\nu)\}$$

¹⁰Hamkins' Gap Forcing Theorem (Theorem 3.7) will not be used in the section, so there is no need to force with $\mathbb{R}^\tau = \text{Add}(\omega, 1) * \mathbb{P}^\tau$.

¹¹Note that there exists a club C disjoint from $\text{supp}(r)$, and $[C]^n \in U^n$. We can therefore assume that $r \cup \{\langle \nu_0, \eta_0 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle\}$ is a condition for a set of $\vec{\nu}$ -s in U^n .

is a dense subset of $\mathbb{P}_{\nu+1}$ above $q \restriction (\nu+1)$. We can assume that $C \cap \text{supp}(q) = \emptyset$ by shrinking C . Then for every $\nu \in C$, there exists a condition $r(\nu) \in \mathbb{P}_\nu$ extending $q \restriction \nu$ such that

$$r(\nu) \cup \{\langle \nu, \eta_0 \rangle\} \cup (q \restriction (\nu+1)) \in e(\nu).$$

Since ${}^\kappa M_U \subseteq M_U$, $j_U(\mathbb{P}) \restriction \kappa = \mathbb{P}$. Moreover, $j_U(q) \restriction \kappa = q$. We therefore deduce that $r = [\nu \mapsto r(\nu)]_U$ is a condition in $\mathbb{P}$ extending q . Furthermore, since $r = j_U(r) \restriction \kappa = [\nu \mapsto r \restriction \nu]_U$ as well, there exists a set $A \in U$ such that for every $\nu \in A$, $r \restriction \nu = r(\nu)$. Then for every $\nu \in A \cap C \in U$,

$$r \restriction \nu \cup \{\langle \nu, \eta_0 \rangle\} \cup (r \restriction (\nu+1)) \in e(\nu).$$

Now proceed by induction. Fix a sequence $\langle \eta_0, \dots, \eta_{n-1}, \eta_n \rangle \in \tau^{n+1}$. Assume that for every $\langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n$, and for every $\nu \in \kappa \setminus \nu_{n-1}$,

$$e(\nu_0, \dots, \nu_{n-1}, \nu) \subseteq \mathbb{P}$$

is a dense open set.

For every $\vec{\nu} = \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n$, let $e(\nu_0, \dots, \nu_{n-1})$ denote the set of conditions r for which there exists a set $A \in U$ such that for every $\nu \in A \setminus (\nu_{n-1} + 1)$,

$$r \cup \{\langle \nu, \eta_n \rangle\} \in e(\nu_0, \dots, \nu_{n-1}, \nu).$$

Claim 4.3. *For every $\vec{\nu} = \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n$, the set $e(\nu_0, \dots, \nu_{n-1})$ is dense open in $\mathbb{P}$.*

Proof. Fix $\vec{\nu} = \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n$. The fact that $e(\nu_0, \dots, \nu_{n-1})$ is open follows, since the set $e(\nu_0, \dots, \nu_{n-1}, \nu_n)$ is open for every $\nu_n \in (\nu_{n-1}, \kappa)$. Thus, let us concentrate on the proof of the density of $e(\nu_0, \dots, \nu_{n-1})$. Fix a condition $p \in \mathbb{P}$. The sequence $\langle e(\vec{\nu}, \nu_n) : \nu_n \in (\nu_{n-1}, \kappa) \rangle$ consists of dense open subsets of $\mathbb{P}$. By the case $n = 1$ in the proof of the Lemma 4.2, there exists $r \geq p$ such that

$$\{\nu_n \in (\nu_{n-1}, \kappa) : r \cup \{\langle \nu_n, \eta_n \rangle\} \in e(\vec{\nu}, \nu_n)\} \in U.$$

It follows that $p \leq r \in e(\nu_0, \dots, \nu_{n-1})$, as desired. $\square$

Fix $p \in \mathbb{P}$. Apply the induction hypothesis to the sequence

$$\langle e(\nu_0, \dots, \nu_{n-1}) : \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n \rangle.$$

By induction, there are $r \geq p$ and $X \in U^n$ such that for every increasing sequence $\vec{\nu} = \langle \nu_0, \dots, \nu_{n-1} \rangle \in X$,

$$r \cup \{\langle \nu_0, \eta_0 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle\} \in e(\vec{\nu}).$$

In particular, for every such sequence $\langle \nu_0, \dots, \nu_{n-1} \rangle$, there exists a set $A(\nu_0, \dots, \nu_{n-1}) \in U$ such that for every $\nu_n \in A(\nu_0, \dots, \nu_{n-1}) \setminus (\nu_{n-1} + 1)$,

$$r \cup \{\langle \nu_0, \eta_0 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle, \langle \nu_n, \eta_n \rangle\} \in e(\nu_0, \dots, \nu_{n-1}, \nu_n).$$

Let

$$Y = \{\langle \nu_0, \dots, \nu_{n-1}, \nu_n \rangle \in [\kappa]^{n+1} : \langle \nu_0, \dots, \nu_{n-1} \rangle \in X \text{ and } \nu_n \in A(\nu_0, \dots, \nu_{n-1})\}.$$

We argue that $Y \in U^{n+1}$. It suffices to prove that $\langle \kappa_0, \dots, \kappa_{n-1}, \kappa_n \rangle \in j_{U^{n+1}}(Y)$. In other words, we need to prove:

1. $\langle \kappa_0, \dots, \kappa_{n-1} \rangle \in j_{U^{n+1}}(X)$. This follows since $X \in U^n$, and thus

$$\langle \kappa_0, \dots, \kappa_{n-1} \rangle \in j_{U^n}(X) \subseteq j_{U^{n+1}}(X).$$

2. $\kappa_n \in j_{U^{n+1}}(\langle \nu_0, \dots, \nu_{n-1} \rangle \mapsto A(\nu_0, \dots, \nu_{n-1}))(\kappa_0, \dots, \kappa_{n-1})$. This follows since $A(\nu_0, \dots, \nu_{n-1}) \in U$ for every $\langle \nu_0, \dots, \nu_{n-1} \rangle$, and thus

$$A^* := j_{U^n}(\langle \nu_0, \dots, \nu_{n-1} \rangle \mapsto A(\nu_0, \dots, \nu_{n-1}))(\kappa_0, \dots, \kappa_{n-1}) \in j_U(U),$$

namely

$$\kappa_n \in j_{j_U(U)}(A^*) = j_{U^{n+1}}(\langle \nu_0, \dots, \nu_{n-1} \rangle \mapsto A(\nu_0, \dots, \nu_{n-1}))(\kappa_0, \dots, \kappa_{n-1}).$$

Thus, $Y \in U^{n+1}$. By shrinking Y in U^{n+1} , we can assume that $Y \subseteq [C]^{n+1}$, where C is some club disjoint from $\text{supp}(r)$. It follows that the condition r has the desired property, witnessed by $Y \in U^{n+1}$. $\square$

Corollary 4.4. *Let U be a normal measure on κ . Fix $n < \omega$ and $\tau < \kappa$. Let $G \subseteq \mathbb{P}$ be generic over V . Then for every sequence $\vec{\eta} = \langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$, the set $U^n \cup \{S_{\vec{\eta}}\}$ generates a κ -complete ultrafilter in $V[G]$, where*

$$S_{\vec{\eta}} = \{\langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n : \forall i < n \ ((\cup G)(\nu_i) = \eta_i)\}.$$

Proof. We construct a $j_{U^n}(\mathbb{P})$ -generic set over M_{U^n} . Let

$$H_{\vec{\eta}} = \{q \in j_{U^n}(\mathbb{P}) : \exists p \in G (q \leq j_{U^n}(p) \cup \{\langle \kappa, \eta_0 \rangle, \dots, \langle \kappa_{n-1}, \eta_{n-1} \rangle\})\}.$$

Let us show that $H_{\vec{\eta}} \subseteq j_{U^n}(\mathbb{P})$ is generic over M_{U^n} .

First, note that for every $m \leq n$ and $p \in \mathbb{P}$, $\kappa_m \notin j_{U^n}(p)$. This is since κ_m belongs to $j_{U^n}(C)$, where $C \subseteq \kappa$ is any club disjoint from the support of p . Thus, $H_{\vec{\eta}}$ consists of legitimate conditions in $j_{U^n}(\mathbb{P})$. Clearly, any pair of conditions in $H_{\vec{\eta}}$ are compatible. Thus, we only need to argue that $H_{\vec{\eta}}$ is $j_{U^n}(\mathbb{P})$ -generic over V . This is an immediate corollary of Lemma 4.2: Given any dense open set $E \subseteq j_{U^n}(\mathbb{P})$ in M_{U^n} , let $\langle e(\nu_0, \dots, \nu_{n-1}) : \langle \nu_0, \dots, \nu_{n-1} \rangle \in [\kappa]^n \rangle$ be a sequence of dense open subsets of $\mathbb{P}$ such that

$$E = j_{U^n}(\langle \nu_0, \dots, \nu_{n-1} \rangle \mapsto e(\nu_0, \dots, \nu_{n-1}))(\kappa_0, \dots, \kappa_{n-1}).$$

Pick a condition $r \in G$ such that

$$\{\vec{\nu} \in [\kappa]^n : r \cup \{\langle \nu_0, \eta_0 \rangle, \dots, \langle \nu_{n-1}, \eta_{n-1} \rangle\} \in e(\nu_0, \dots, \nu_{n-1})\} \in U^n.$$

Then $j_{U^n}(r) \cup \{\langle \kappa, \eta_0 \rangle, \dots, \langle \kappa_{n-1}, \eta_{n-1} \rangle\} \in E \cap H_{\vec{\eta}}$, as desired. $\square$

As a corollary, we obtain a complete characterization of all possible lifts of U^n in $V[G]$.

Corollary 4.5. *Let U be a normal measure on κ . Fix $n < \omega$ and $\tau < \kappa$. Let $G \subseteq \mathbb{P}$ be generic over V . For every σ -complete ultrafilter $W \in V[G]$ such that $U^n \subseteq W$, there exists $\vec{\eta} = \langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$ such that W is the measure generated by $U^n \cup \{S_{\vec{\eta}}\}$.*

Proof. Let $j_W : V[G] \rightarrow M[H] = M[j_W(G)]$ be the ultrapower embedding associated with W .

Since $U^n \subseteq W$, W concentrates on the set $[I]^n$ consisting of increasing sequences of inaccessible cardinals (in V) of length n . Thus, $[id]_W = \{\lambda_0, \dots, \lambda_{n-1}\}$ for an increasing sequence $\lambda_0 < \dots < \lambda_{n-1}$ of inaccessible cardinals in M .

It therefore follows that the generic $H = j_W(G)$ assigns to each λ_i (for $0 \leq i \leq n-1$) some $\eta_i = (\cup H)(\lambda_i) < j_W(\tau) = \tau$. Thus, letting $\vec{\eta} = \langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$, we deduce that $S_{\vec{\eta}} \in W$, and W is the measure generated by $U^n \cup \{S_{\vec{\eta}}\}$. $\square$

All the results provided so far have not required any assumptions on the ground model V . However, to obtain a full classification of all σ -complete ultrafilters in $V[G]$, further assumptions, such as the assumption that $V = L[U]$, will be necessary. Recall that the model $L[U]$, introduced by Kunen in his seminal work [20], is the minimal canonical inner model containing a measurable cardinal. The following properties are well-known:

Theorem 4.6. *Assume that κ is measurable and U is a normal measure on κ . Then there exists a canonical inner model $L[U]$ which satisfies the following properties:*

1. (Kunen, [20, Corollary 6.5]) κ is the unique measurable cardinal in $L[U]$, κ carries a unique normal measure $\bar{U} = U \cap L[U]$ in $L[U]$, and $L[U] = L[\bar{U}]$.
2. (Kunen, [20, Theorem 7.6]) For every σ -complete ultrafilter $W \in L[U]$, there exists $n < \omega$ such that $W \equiv_{RK} (\bar{U})^n$.
3. (Silver, [25]) $L[U] \models \text{GCH}$.

We now proceed to the proof of Theorem 1.4.

Proof of Theorem 1.4. Assume $V = L[U]$. By a mild abuse of notation, we will identify the measure U with its restriction $\bar{U} = U \cap L[U] \in L[U]$.

Let $G \subseteq \mathbb{P}$ be generic over V . In order to simplify the notation, let $\mathbb{P} = \mathbb{P}^\tau$.

The proof will rely heavily on the fact that every σ -complete ultrafilter $W \in V[G]$ on an ordinal χ satisfies $W \cap V \in V$. Indeed, the restriction $j_W \upharpoonright V$ is an iterated ultrapower j_{U^m} of U over $V = L[U]$ for some $m < \omega$.¹² Consequently,

$$W \cap L[U] = \{X \subseteq \chi : [id]_W \in j_{U^m}(X)\} \in L[U].$$

¹²That $j_W \upharpoonright L[U]$ is an iterated ultrapower of $L[U]$ is a folklore consequence of Kunen's work [20]. Since $\mathbb{P}$ adds no new ω -sequences, this iteration must be finite.

We begin by proving clause 1 of Theorem 1.4. By Lemma 3.6, the measure U admits τ distinct lifts $\langle U_\eta^*: \eta < \tau \rangle$ to normal measures on κ in $V[G]$. It therefore remains to show that every normal measure $W \in V[G]$ on κ arises as one of these lifts. To this end, let $W \in V[G]$ be a normal measure on κ . By the previous paragraph, $W \cap V \in V$, and $W \cap V$ is a normal measure in V as it inherits its normality from W . By clause 1 of Theorem 4.6, it follows that $W \cap L[U] = U$. Now let $\eta = j_W(\cup G)(\kappa)$. Then $\kappa \in j_W(S_\eta)$, namely $S_\eta \in W$. Since $U \cup \{S_\eta\}$ generates the measure U_η^* and $U \cup \{S_\eta\} \subseteq W$, we deduce that $W = U_\eta^*$. The fact that all the measures U_η^* for $\eta < \tau$ have the same ultrapower was already established in Lemma 3.6.

Let us proceed and prove clauses 2 and 3 of Theorem 1.4. Let $W \in V[G]$ be an arbitrary σ -complete ultrafilter in $V[G]$, concentrating on an ordinal χ . By the previous paragraphs, $W \cap L[U]$ is a σ -complete ultrafilter in $L[U]$. By clause 2 of Theorem 4.6, $W \cap L[U]$ is Rudin-Keisler equivalent to U^n for some $n < \omega$. Fix a set $Y \in W$ and an injection $h: Y \rightarrow [\kappa]^n$ witnessing this (namely, for every set $X \subseteq Y$, $X \in W \cap L[U]$ if and only if $h[X] \in U^n$). By replacing W with $h_*(W) = \{X \subseteq [\kappa]^n: h^{-1}[X] \in W\}$, we can assume from now on that W concentrates on $[\kappa]^n$ and $W \supseteq U^n$. This is harmless, since our goal is characterizing j_W and $\text{Ult}(L[U][G], W)$, which are invariant under Rudin-Keisler equivalence. By Corollary 4.5, there exists $\vec{\eta} = \langle \eta_0, \dots, \eta_{n-1} \rangle \in \tau^n$ such that W is the measure generated by $U^n \cup \{S_{\vec{\eta}}\}$.

We proceed and prove that $j_W^{V[G]}: V[G] \rightarrow \text{Ult}(V[G], W)$ is a finite iterated ultrapower of $V[G]$ via normal measures.

Let $W^0 \in V[G]$ be the normal measure on κ generated by $U \cup \{S_{\eta_0}\}$ (in the notation of Lemma 3.6). Let $N^0 \simeq \text{Ult}(V[G], W^0)$, and note that $N^0 = M_U[H_{\eta_0}]$, where $H_{\eta_0} = j_{W^0}(G)$ is the $j_U(\mathbb{P})$ -generic generated by $(\cup j_U[G]) \cup \{\langle \kappa, \eta_0 \rangle\}$.

Over $N^0 = M_U[H_{\eta_0}]$, let W^1 be the normal measure on κ_1 generated by $j_U(U) \cup \{S_{\eta_1}^1\}$, where $S_{\eta_1}^1 = \{\alpha < \kappa_1: (\cup H_{\eta_0})(\alpha) = \eta_1\}$. Set $N^1 \simeq \text{Ult}(N^0, W^1)$. We argue that N^1 has the form $M_{U^2}[H_{\langle \eta_0, \eta_1 \rangle}]$, where $H_{\langle \eta_0, \eta_1 \rangle} = j_{W^1}(H_{\eta_0})$ is the $j_{U^2}(\mathbb{P})$ -generic generated by $\{(\cup j_{U^2}[G]) \cup \{\langle \kappa, \eta_0 \rangle, \langle \kappa_1, \eta_1 \rangle\}\}$.

Indeed, inside $V[G]$, the ground of $\text{Ult}(V[G], \mathcal{U})$ for every normal measure $\mathcal{U} \in V[G]$ is definable as $\text{Ult}(V, \mathcal{U} \cap V)$ (recall that V itself is definable in $V[G]$ by Laver's Ground Model Definability Theorem). Furthermore, $j_{\mathcal{U}}(G)$ is the generic set generated by $(\cup j_{\mathcal{U} \cap V}[G]) \cup \{\langle \kappa, \eta \rangle\}$, where η is the unique ordinal below τ such that $S_\eta \in \mathcal{U}$. Keeping all of this in mind, we can now use elementarity to deduce what $N^1 \simeq \text{Ult}(N^0, W^1)$ looks like inside $N^0 = M_U[H_{\eta_0}]$: The ground of $\text{Ult}(N^0, W^1)$ is

$$\text{Ult}(M_U, W^1 \cap M_U) = \text{Ult}(M_U, j_U(U)) = M_{U^2},$$

and the corresponding generic set $j_{W^1}(H_{\eta_0})$ is the one generated by

$$(\cup j_{W^1 \cap M_U}[H_{\eta_0}]) \cup (\{\langle \kappa_1, \eta_1 \rangle\}).$$

Since $W^1 \cap M_U = j_U(U)$ and H_{η_0} itself is generated by $j_U[G] \cup \{\langle \kappa, \eta_0 \rangle\}$, we

deduce that $j_{W^1}(H_{\eta_0})$ is the generic generated by

$$(\cup j_{U^2}[G]) \cup (\{\langle \kappa, \eta_0 \rangle, \langle \kappa_1, \eta_1 \rangle\},$$

which is exactly the generic $H_{\langle \eta_0, \eta_1 \rangle}$ defined above.

Continuing this inductively, we define models N^i , normal measures $W^i \in N^i$, ordinals $\eta_i < \tau$, and generics $H_{\langle \eta_0, \dots, \eta_i \rangle} \subseteq j_{U^i}(\mathbb{P})$, such that for every $1 \leq i \leq n-1$,

1. $N^i = \text{Ult}(N^{i-1}, W^i) \simeq M_{U^{i+1}}[H_{\langle \eta_0, \dots, \eta_i \rangle}]$.
2. W^{i+1} is generated inside N^i from $j_{U^{i+1}}(U) \cup \{S_{\eta_{i+1}}^{i+1}\}$, where $S_{\eta_{i+1}}^{i+1} = \{\alpha < \kappa_{i+1} : (\cup H_{\langle \eta_0, \dots, \eta_i \rangle})(\alpha) = \eta_{i+1}\}$.
3. $H_{\langle \eta_0, \dots, \eta_{i+1} \rangle} \subseteq j_{U^{i+2}}(\mathbb{P})$ is generated by

$$(\cup j_{W^{i+1} \cap M_{U^{i+1}}}[H_{\langle \eta_0, \dots, \eta_i \rangle}]) \cup \{\langle \kappa_{i+1}, \eta_{i+1} \rangle\}.$$

The measure $W^{i+1} \cap M_{U^{i+1}}$ is equal to $j_{U^{i+1}}(U)$. In particular, $H_{\langle \eta_0, \dots, \eta_{i+1} \rangle}$ is generated by

$$(\cup j_{U^{i+2}}[G]) \cup \{\langle \kappa, \eta_0 \rangle, \dots, \langle \kappa_{i+1}, \eta_{i+1} \rangle\}.$$

Note that $H_{\langle \eta_0, \dots, \eta_{n-1} \rangle}$ is the same generic from the proof of Corollary 4.4, corresponding to the ultrapower embedding associated with the measure generated by $U^n \cup \{S_{\langle \eta_0, \dots, \eta_{n-1} \rangle}\}$, which we have already identified as W . It follows that $N^{n-1} = M_{U^n}[H_{\langle \eta_0, \dots, \eta_{n-1} \rangle}]$ is the transitive collapse of $\text{Ult}(V[G], W)$. In other words, the ultrapower with W is the result of the successive ultrapowers taken with the normal measures $W^0, W^1, \dots, W^{n-1}$ described above. This proves clause 2 of Theorem 1.4.

Finally, note that for every $i \leq n-1$, $N^i \simeq \text{Ult}(V[G], (U_0^*)^{i+1})$ (where U_0^* is as in Lemma 3.6 for $\eta = 0$). This follows by induction, using the fact that any pair of normal measures in $V[G]$ have the same ultrapower. Indeed, for $i = 0$, $N^0 \simeq \text{Ult}(V[G], W^0) \simeq \text{Ult}(V[G], U_0^*)$. Assuming the statement for i , note that $N^{i+1} \simeq \text{Ult}(N^i, W^{i+1})$. But by the induction hypothesis, $N^i \simeq \text{Ult}(V[G], (U_0^*)^{i+1})$, and $j_{(U_0^*)^{i+1}}(U_0^*)$ is a normal measure on κ_{i+1} in N^i . It follows that $N^{i+1} \simeq \text{Ult}(N^i, W^{i+1}) \simeq \text{Ult}(N^i, j_{(U_0^*)^{i+1}}(U_0^*)) \simeq \text{Ult}(V[G], (U_0^*)^{i+2})$.

In particular, the above inductive argument shows that N^{n-1} is isomorphic to $\text{Ult}(V[G], (U_0^*)^n)$. This proves clause 3 of Theorem 1.4. $\square$

Proof of Theorem 1.5. We argue that Weak UA + $\neg$ UA is consistent relative to the existence of a measurable cardinal. Assume the latter, and let $V = L[U]$ be the canonical inner model containing a measurable cardinal κ and a unique normal measure U on κ .

Let $G \subseteq \mathbb{P}^\tau$ be generic over V , where τ is any cardinal such that $2 \leq \tau < \kappa$. By Theorem 1.4, κ carries two distinct normal measures in $V[G]$ which are Mitchell incomparable (since all normal measures in $V[G]$ have the same ultrapower, and no measure can belong to its own ultrapower). By a theorem

of Goldberg, UA implies the linearity of the Mitchell order (see [14, Theorem 2.3.11]). In particular, UA does not hold in $V[G]$.

It remains to show that $V[G] \models \text{Weak UA}$. Let W, W' be a pair of σ -complete ultrafilters in $V[G]$. It suffices to prove that $N \simeq \text{Ult}(V[G], W)$ and $N' \simeq \text{Ult}(V[G], W')$ have common internal ultrapowers. By Theorem 1.4, there are $n, n' < \omega$ such that $N \simeq \text{Ult}(V[G], (U_0^*)^n)$ and $N' \simeq \text{Ult}(V[G], (U_0^*)^{n'})$. If $n = n'$, $N = N'$, and we are done. Assume otherwise, and without loss of generality, assume that $n < n'$. In N , consider the measure $\mathcal{V} = (j_W(U_0^*))^{n'-n}$. Since $N \simeq \text{Ult}(V[G], (U_0^*)^n)$,

$$\text{Ult}(N, \mathcal{V}) \simeq \text{Ult}(V[G], (U_0^*)^{n'}),$$

namely $N' \simeq \text{Ult}(N, \mathcal{V})$, as desired. $\square$

The Goldberg–Benhamou model of Weak UA + $\neg$ UA (see [9]) has a unique measurable cardinal and a single normal measure. In contrast, Theorem 1.5 establishes the consistency of Weak UA + $\neg$ UA with the existence of multiple normal measures on the unique measurable cardinal.

5 Friedman–Magidor for extenders

In the previous sections, our goal was to classify lifts of normal measures (and their iterations) in forcing extensions via the splitting forcing $\mathbb{P}^\tau$. We now turn to lifts of extenders.

We will rely on the standard definition of (κ, λ) -extenders for $\lambda > \kappa$ (see [19, chapter 5, section 26]). We view each such extender as a sequence of measures $E = \langle E_a : a \in [\lambda]^{<\omega} \rangle$, where E_a is a measure on the set $[\kappa]^{|a|}$. As usual, we can form the ultrapower $M_E \simeq \text{Ult}(V, E)$ with respect to E , and we denote by $j_E : V \rightarrow M_E$ the associated ultrapower embedding. We say that a (κ, λ) -extender E is κ -closed if $V \models {}^\kappa M_E \subseteq M_E$. Throughout this section, we say that two extenders E and E' are *equivalent* if $j_E = j_{E'}$.

As before, let I be the set of inaccessible cardinals below κ .

For the remainder of the section, fix a function $g : \kappa \rightarrow \kappa$ such that, for every $\alpha < \kappa$, $g(\alpha) < \min(I \setminus (\alpha + 1))$. We will consider in this section κ -closed (κ, λ) -extenders E such that $j_E(g)(\kappa)$ bounds the set of generators of j_E (in particular, E is equivalent to the $(\kappa, j_E(g)(\kappa))$ -extender derived from it). Let us recall the definition of a generator of an elementary embedding.

Definition 5.1. Assume that $j : V \rightarrow M$ is an elementary embedding. We say that an ordinal η is a *generator* of j if there is no $a \in [\eta]^{<\omega}$ and function $f \in V$ such that $\eta = j(f)(a)$.

The main forcing considered in this section will be the following generalization of the splitting forcing from Definition 3.2.

Definition 5.2. Let $g: \kappa \rightarrow \kappa$ and I be as above. Assume that $h: \kappa \rightarrow \kappa$ is a function such that, for every $\alpha < \kappa$, $h(\alpha) \leq g(\alpha)$. Let

$$\mathbb{P}^h = \{f \in \prod_{\alpha \in X} h(\alpha) : X = \text{dom}(f) \text{ is a nowhere stationary subset of } I\}.$$

As usual, we order $\mathbb{P}^h$ by inclusion, and define $\mathbb{R}^h = \text{Add}(\omega, 1) * \mathbb{P}^h$.

Remark 5.3. Let us elaborate on the role of the functions g, h in this section.

1. The function g describes the class of extenders E to which our methods apply. As mentioned in the paragraph preceding Definition 5.1, our methods provide control over all lifts of κ -closed (κ, λ) -extenders E such that $j_E(g)(\kappa)$ bounds all the generators of E . We conjecture that these methods can be generalized to a wider class of extenders (see Remark 5.6).
2. The function h determines the number of nonequivalent lifts of a given ground model extender E to the generic extension. More specifically, such an extender E will lift in $|j_E(h)(\kappa)|$ -nonequivalent ways.

Throughout this section, we fix the functions g and h and adopt the notation $\mathbb{P} = \mathbb{P}^h$ and $\mathbb{R} = \mathbb{R}^h$. Specific instances of g, h will be considered only in the proof of Theorem 1.6, where they will be specified explicitly.

Our forcing $\mathbb{P}$ can be viewed as a nonstationary support product $\prod_{\alpha \in I}^{NS} \mathbb{Q}_\alpha$, where $\mathbb{Q}_\alpha = \{0_{\mathbb{Q}_\alpha}\} \cup h(\alpha)$ is an atomic forcing, in which $h(\alpha) \subseteq \mathbb{Q}_\alpha$ is a maximal antichain. The requirement that each $g(\alpha)$ be below $\min(I \setminus (\alpha + 1))$ ensures that $\mathbb{P}$ satisfies the assumptions of the Fusion Lemma 2.3. As in Sections 2 and 3, GCH ensures that $\mathbb{P}$ and $\mathbb{R}$ preserve cardinals.

Our main goal in this section will be to prove the following theorem.

Theorem 5.4. *Assume GCH, and let $g, h: \kappa \rightarrow \kappa$, $\mathbb{P}$, and $\mathbb{R}$ be as above. Let $G \subseteq \mathbb{R}$ be generic over V . Then the following properties hold:*

1. *For every κ -closed (κ, λ) -extender E in V whose generators are all below $j_E(g)(\kappa)$, and for each $\eta < j_E(h)(\kappa)$, there exists a κ -closed (κ, λ) -extender $E_\eta^* \in V[G]$ such that $j_{E_\eta^*} \upharpoonright V = j_E$. Furthermore, all the extenders $\langle E_\eta^* : \eta < j_E(h)(\kappa) \rangle$ are nonequivalent and have the same ultrapower.*
2. *Every κ -closed (κ, λ) -extender $E^* \in V[G]$ whose generators all lie below $j_{E^*}(g)(\kappa)$ and with $\lambda \geq j_{E^*}(g)(\kappa)$ is, up to equivalence, of the form E_η^* , for some κ -closed (κ, λ) -extender $E \in V$, where $\eta < j_E(h)(\kappa)$ and the generators of E all lie below $j_E(g)(\kappa)$.*

Remark 5.5. The requirement $\lambda \geq j_{E^*}(g)(\kappa)$ in clause 2 above is necessary in order to ensure that $j_{E^*} \upharpoonright V$ is a (κ, λ) -extender in V . In fact, Remark 5.9 below shows that it is possible that for some normal measure $W \in V[G]$ and extender $E \in V$, $W = E_\eta^*$ where $\eta > \kappa + 1$ is a generator of j_E . In other words, even a $(\kappa, \kappa + 1)$ -extender $W \in V[G]$ might satisfy that $j_W \upharpoonright V = j_E$ where E is not a $(\kappa, \kappa + 1)$ -extender.

Remark 5.6. Recall that the requirement that $g(\alpha) < \min(I \setminus (\alpha + 1))$ for all $\alpha < \kappa$ was imposed to ensure that the Fusion Lemma 2.3 applies to $\mathbb{P}$. We conjecture that this requirement can be relaxed by “spacing out” the product, namely by forcing with $\mathbb{P} = \prod_{\alpha \in J}^{NS} \mathbb{Q}_\alpha$, where J is a sufficiently sparse stationary subset of I . In this setting, the requirement $g(\alpha) < \min(J \setminus (\alpha + 1))$ is less restrictive, and Theorem 5.4 should apply to a broader class of extenders.

For example, let $J \subseteq I$ be the set of inaccessible cardinals that are limits of inaccessible cardinals, and define $g: \kappa \rightarrow \kappa$ by $g(\alpha) = \min(I \setminus (\alpha + 1))$. Then g satisfies $g(\alpha) < \min(J \setminus (\alpha + 1))$ for all $\alpha < \kappa$. We conjecture that the methods developed in this section can be used to classify the lifts of extenders E for which the generators of j_E are all below $j_E(g)(\kappa)$. This includes, in particular, extenders E whose generators form an unbounded subset of $\min(j_E(I) \setminus (\kappa + 1))$. We do not pursue this direction further in the present paper.

Lemma 5.7. *Assume that E is a κ -closed (κ, λ) -extender in V whose generators all lie below $j_E(g)(\kappa)$. Let $G \subseteq \mathbb{R}$ be generic over V . Then in $V[G]$, there are $|j_E(h)(\kappa)|$ nonequivalent κ -closed (κ, λ) -extenders $\langle E_\eta^*: \eta < j_E(h)(\kappa) \rangle \in V[G]$, all with the same ultrapower, such that each $j_{E_\eta^*}$ is a lift of j_E .*

Proof. Write $G = g' * G'$, where $g' \subseteq \text{Add}(\omega, 1)$ is generic over V , and $G' \subseteq \mathbb{P}$ is generic over $V[g']$. The extender E naturally and uniquely lifts to an extender $E' \in V[g']$, namely the (κ, λ) -extender derived from j' , the standard lift of j to an embedding from $V[g']$ into $M_E[g']$. Our aim is to show that E' lifts in (at least) $|j_{E'}(h)(\kappa)| = |j_E(h)(\kappa)|$ nonequivalent ways in the extension $V[g'][G']$.

To establish this, it suffices to prove that over any ground model V , whenever $G \subseteq \mathbb{P}$ is generic over V and $E \in V$ is a (κ, λ) -extender, there are at least $|j_E(h)(\kappa)|$ nonequivalent κ -closed (κ, λ) -extenders $\langle E_\eta^*: \eta < j_E(h)(\kappa) \rangle$, all of them with the same ultrapower, such that each $j_{E_\eta^*}$ lifts j_E .

Thus, we change the assumptions to the ones mentioned in the previous paragraph. Fix $\eta < j_E(h)(\kappa)$. Define, in $V[G]$,

$$H_\eta = \{q \in j_E(\mathbb{P}) : \exists p \in G \ (q \leq j_E(p) \cup \{\langle \kappa, \eta \rangle\})\}.$$

We will argue that H_η is $j_E(\mathbb{P})$ -generic over V . The crux of the argument lies in proving that H_η meets every dense open subset of $j_E(\mathbb{P})$ present in M_E .

Fix a dense open subset $D \in M_E$ of $j_E(\mathbb{P})$. Let $a \in [j_E(g)(\kappa)]^{<\omega}$ be a set of generators and $e: [\kappa]^{|a|} \rightarrow V$ be a function such that $D = j_E(e)(a)$. Define, for every $\alpha < \kappa$,

$$d(\alpha) = \{q \in \mathbb{P} : q \restriction (\alpha + 1) \Vdash \forall b \in [g(\alpha)]^{|a|} \exists r \in \dot{G}_{\alpha+1} (r \restriction (q \setminus (\alpha + 1)) \in e(b))\}.$$

Let us argue that $d(\alpha) \subseteq \mathbb{P}$ is dense open for every $\alpha < \kappa$. Assume that $q \in \mathbb{P}$ is given. For every $b \in [g(\alpha)]^{|a|}$, consider the set

$$d_b(\alpha) = \{s \in \mathbb{P} \setminus (\alpha + 1) : \forall r \in \mathbb{P}_{\alpha+1} \exists r' \in \mathbb{P}_{\alpha+1} \ (r' \geq r \text{ and } r' \restriction s \in e(b))\}.$$

Each set $d_b(\alpha)$ is forced to be a dense open subset of $\mathbb{P} \setminus (\alpha + 1)$, since $e(b)$ is dense open, and the poset $\mathbb{P} \setminus (\alpha + 1)$ is more than $|\mathbb{P}_{\alpha+1}|$ -closed (note that

$|\mathbb{P}_{\alpha+1}| = \max\{2^{|\alpha|}, |h(\alpha)|\} < \min(I \setminus (\alpha + 1))$. Since $\mathbb{P} \setminus (\alpha + 1)$ is more than $|g(\alpha)|$ -distributive, there exists a condition $s \in \mathbb{P} \setminus (\alpha + 1)$ such that

$$s \in \bigcap_{b \in [g(\alpha)]^{<\omega}} d_b(\alpha).$$

Let $q^* = (q \restriction (\alpha + 1)) \smallfrown s$. Then $q \leq q^* \in d(\alpha)$, as desired.

Next, apply the Fusion Lemma 2.3 on the sequence of dense open sets $\langle d(\alpha) : \alpha < \kappa \rangle$. The lemma produces a condition $p \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$,

$$\{q \in \mathbb{P}_{\alpha+1} : q \smallfrown (p \setminus (\alpha + 1)) \in d(\alpha)\}$$

is dense open above $p \restriction (\alpha + 1)$. In particular, for every such $\alpha \in C$,

$$p \restriction (\alpha + 1) \Vdash \forall b \in [g(\alpha)]^{|\alpha|} \exists r \in \dot{G}_{\alpha+1} (r \smallfrown (p \setminus (\alpha + 1))) \in e(b).$$

Since $C \subseteq \kappa$ is a club, $\kappa \in j_E(C)$, and thus, over M_E ,

$$p \smallfrown 0_{\mathbb{Q}_\kappa} \Vdash \forall b \in [j_E(g)(\kappa)]^{<\omega} \exists r \in j_E(\dot{G})_{\kappa+1} (r \smallfrown (j_E(p) \setminus (\kappa + 1))) \in j_E(e)(b),$$

where $\mathbb{Q}_\kappa := j_E(\langle \mathbb{Q}_\alpha : \alpha < \kappa \rangle)(\kappa)$. Since $a \in [j_E(g)(\kappa)]^{<\omega}$, we deduce that

$$p \smallfrown 0_{\mathbb{Q}_\kappa} \Vdash \exists r \in j_E(\dot{G})_{\kappa+1} (r \smallfrown (j_E(p) \setminus (\kappa + 1))) \in j_E(e)(a) = D).$$

We already have at our disposal a natural $j_E(\mathbb{P})_{\kappa+1}$ -generic set over M_E . Indeed, since E is κ -closed, $j_E(\mathbb{P})_\kappa = \mathbb{P}$, and thus G is $j_E(\mathbb{P})_\kappa$ -generic over M_E . Furthermore, η dictates a $\mathbb{Q}_\kappa$ -generic over $M_E[G]$, because $\eta < j_E(h)(\kappa)$. Since $p \smallfrown 0_{\mathbb{Q}_\kappa}$ belongs to the $j_E(\mathbb{P})_{\kappa+1}$ -generic set just described, any statement being forced by it holds in $M_E[G]$. Thus, for some $r \in G$,

$$r \smallfrown \{\langle \kappa, \eta \rangle\} \smallfrown (j_E(p) \setminus (\kappa + 1)) \in D.$$

Finally, by extending p inside G so it extends r , we achieve that $j_E(p) \cup \{\langle \kappa, \eta \rangle\} \in D \cap H_\eta$.

This proves that each H_η is $j_E(\mathbb{P})$ -generic over M_E . Since $j_E[G] \subseteq H_\eta$, j_E lifts to an embedding $j_{E,\eta}^* : V[G] \rightarrow M_E[H_\eta]$. By standard arguments, $j_{E,\eta}^*$ is the ultrapower embedding of the (κ, λ) -extender E_η^* derived from it. The fact that E_η^* is κ -closed follows from the following claim.

Claim 5.8. $V[G] \models {}^\kappa M_E[H_\eta] \subseteq M_E[H_\eta]$.

Proof. It suffices to prove that $V[G] \models {}^\kappa \text{Ord} \subseteq M_E[G]$. Let $f \in V[G]$, $f : \kappa \rightarrow \text{Ord}$ be given. Fix a $\mathbb{P}$ -name $\dot{f} \in V$ for f , and assume that the weakest condition in $\mathbb{P}$ forces that $\dot{f}$ is a sequence of ordinals of length κ . For every $\alpha < \kappa$, let

$$e(\alpha) = \{q \in \mathbb{P} : q \restriction (\alpha + 1) \Vdash \exists g \in {}^\alpha \text{Ord} \left(q \setminus (\alpha + 1) \Vdash \dot{f} \restriction \alpha = \check{g} \right)\}.$$

Each set $e(\alpha)$ for $\alpha < \kappa$ can be easily seen to be dense open, because the forcing $\mathbb{P} \setminus (\alpha + 1)$ is more than $|\alpha|$ -closed. (Indeed, for every condition $q \in \mathbb{P}$, $q \setminus (\alpha + 1)$

can be extended to a condition that decides all the values $\langle \dot{f}(\beta) : \beta < \alpha \rangle$, by closure). By the Fusion Lemma 2.3, there exist $p \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$,

$$p \restriction (\alpha + 1) \Vdash \exists g_\alpha \in {}^\alpha \text{Ord} \left(p \restriction (\alpha + 1) \Vdash \dot{f} \restriction \alpha = \check{g}_\alpha \right).$$

Let $\dot{g}_\alpha \in V_\kappa$ be a $\mathbb{P}_{\alpha+1}$ -name for g_α as above. Since $V \models {}^\kappa M_E \subseteq M_E$, $\langle \dot{g}_\alpha : \alpha \in C \rangle \in M_E$. In particular, $f \in M_E[G]$, because f can be computed in $M_E[G]$ from $\langle \dot{g}_\alpha : \alpha \in C \rangle$ and G . $\square$

Finally, the extenders $\langle E_\eta^* : \eta < j_E(h)(\kappa) \rangle$ are nonequivalent, since their elementary embeddings disagree on their value at G . Nevertheless, for every $\eta \neq \eta'$ below $j_E(h)(\kappa)$, $M_E[H_\eta] = M_E[H_{\eta'}]$, so $E_\eta^*, E_{\eta'}^*$ have the same ultrapower. Indeed, $H_{\eta'}$ can be easily computed inside $M_E[H_\eta]$ by changing the generic value of H_η at coordinate κ from η to η' .

This concludes the proof of Lemma 5.7. $\square$

Remark 5.9. Note that η is not a generator of $j_{E,\eta}^* = j_{E_\eta^*}$, even if η was a generator of E , since $\eta = j_{E_\eta^*}(\cup G)(\kappa)$ in $V[G]$. In fact, this shows that $\mathbb{P}$ can be used to “kill” a generator of an extender $E \in V$ while lifting the extender.

This observation shows that the classification of normal measures $W \in V[G]$ on κ is more complicated than the one exhibited in Theorem 1.2. For example, let us force with $\mathbb{P}$ over a ground model V which contains a κ -complete ultrafilter $F \in V$ such that j_F has two distinct generators, κ and ζ , where $\zeta \in (\kappa, j_F(h)(\kappa))$. By Lemma 5.7, j_F lifts to an extender embedding $j_{F,\eta}^*$ whose only generator is κ . In particular, $j_{F,\eta}^* = j_W^{V[G]}$, where $W \in V[G]$ is the normal measure on κ derived from $j_{F,\eta}^*$ using κ as a seed. Note that $j_F \subseteq j_W^{V[G]}$, and $W \cap V$ is the normal measure defined in V by $U = \{X \subseteq \kappa : \kappa \in j_F(X)\}$. In particular, $W \supseteq U$, but $j_W^{V[G]}$ doesn't lift j_U . This contrasts with the situation exhibited in Theorem 1.2, in which for every normal measure $W \in V[G]$, $j_W^{V[G]} \supseteq j_{W \cap V}$.

Theorem 5.10. *Assume GCH. Let $G \subseteq \mathbb{R}$ be generic over V . Let E^* be a κ -closed (κ, λ) -extender of $V[G]$ whose generators lie below $j_{E^*}(g)(\kappa)$, and assume that $\lambda \geq j_{E^*}(g)(\kappa)$. Then there exists a κ -closed (κ, λ) -extender $E \in V$ with generators below $j_E(g)(\kappa)$ and $\eta < j_E(h)(\kappa)$ such that E^* is equivalent to E_η^* .*

Proof. Let $M[H] \simeq \text{Ult}(V[G], E^*)$. By Hamkins' Gap Forcing Theorem 3.7, $j_{E^*} \restriction V : V \rightarrow M$ is definable in V . Let E be the (κ, λ) -extender derived from $j_{E^*} \restriction V$ in V .

We argue first that $j_E = j_{E^*} \restriction V$ and $M = M_E$. Define $k : M_E \rightarrow M$ such that, for every $a \in [\lambda]^{<\omega}$ and $f : [\kappa]^{|a|} \rightarrow V$ in V , $k(j_E(f)(a)) = j_{E^*}(f)(a)$.

Claim 5.11. *k is elementary and $j_{E^*} \restriction V = k \circ j_E$.*

Proof. Let $a, b \in [\lambda]^{<\omega}$, and assume that $f_0: [\kappa]^{|a|} \rightarrow V$, $f_1: [\kappa]^{|b|} \rightarrow V$ are functions in V . We argue that

$$j_E(f_0)(a) = j_E(f_1)(b) \iff j_{E^*}(f_0)(a) = j_{E^*}(f_1)(b).$$

Recall the following standard notation: For $x \subseteq y$ in $[\lambda]^{<\omega}$, denote by $\pi_{y,x}: [\kappa]^{|y|} \rightarrow [\kappa]^{|x|}$ the standard projection,

$$\pi_{y,x}(\langle \nu_0, \dots, \nu_{|y|-1} \rangle) = \langle \nu_{i_0}, \dots, \nu_{i_{|x|-1}} \rangle,$$

where $\langle i_0, \dots, i_{|x|-1} \rangle$ is an increasing sequence indicating the coordinates of x inside y , namely $x(j) = y(i_j)$ for every $j < |x|$.

Given f_0, f_1, a, b as above, let $c = b \cup a$. Then

$$\begin{aligned} j_E(f_0)(a) = j_E(f_1)(b) &\iff j_E(f_0 \circ \pi_{c,a})(c) = j_E(f_1 \circ \pi_{c,b})(c) \\ &\iff \{\vec{\nu} \in [\kappa]^{|c|}: (f_0 \circ \pi_{c,a})(\vec{\nu}) = (f_1 \circ \pi_{c,b})(\vec{\nu})\} \in E_c \\ &\iff c \in j_{E^*} \left(\{\vec{\nu} \in [\kappa]^{|c|}: (f_0 \circ \pi_{c,a})(\vec{\nu}) = (f_1 \circ \pi_{c,b})(\vec{\nu})\} \right) \\ &\iff j_{E^*}(f_0 \circ \pi_{c,a})(c) = j_{E^*}(f_1 \circ \pi_{c,b})(c) \\ &\iff j_{E^*}(f_0)(a) = j_{E^*}(f_1)(b). \end{aligned}$$

Building on this argument, the full elementarity of k can be shown inductively. Finally, the fact that $j_{E^*} \upharpoonright V = k \circ j_E$ can be easily shown from the definition of k . $\square$

Let us argue that k is the identity map. By Claim 5.11, this ensures that $j_{E^*} \upharpoonright V = j_E$ and $M = M_E$.

Note that $\text{crit}(k) \geq \lambda$, since E was defined as the (κ, λ) -extender derived from $j_{E^*} \upharpoonright V$. In fact, the critical point of k is the least generator (if such exists) of $j_{E^*} \upharpoonright V$ which is greater than or equal to λ . We argue that no such generator exists. In fact, we argue that there is no generator ζ of E such that $\zeta \geq j_E(g)(\kappa)$.

Fix $\zeta \geq j_E(g)(\kappa)$, and write, in $V[G]$, $\zeta = j_{E^*}(f)(a)$ for some $a \in [j_E(g)(\kappa)]^{<\omega}$ and $f: [\kappa]^{|a|} \rightarrow \text{Ord}$ in $V[G]$. Our goal is to prove that ζ is not a generator of $j_{E^*} \upharpoonright V$. We would like to use here the following strengthening of Corollary 2.4.

Claim 5.12. *There exist a partial function $F \in V$ and a club $C \subseteq \kappa$ such that, for every $\alpha \in C$ and $\vec{\nu} \in [g(\alpha)]^{<\omega}$,*

$$f(\vec{\nu}) \in F(\alpha, \vec{\nu}) \text{ and } |F(\alpha, \vec{\nu})| \leq \max\{\alpha^+, |h(\alpha)|\}.$$

Proof. For every $\alpha < \kappa$, define

$$d(\alpha) = \{q \in \mathbb{P}: q \upharpoonright (\alpha+1) \Vdash \forall \vec{\nu} \in [g(\alpha)]^{<\omega} \exists \gamma_\nu \in \text{Ord} (q \restriction (\alpha+1) \Vdash \dot{f}(\vec{\nu}) = \check{\gamma}_\nu)\}.$$

Each $d(\alpha)$ is dense open, since $\mathbb{P} \setminus (\alpha+1)$ is more than $|g(\alpha)|$ -closed. (Indeed, for every condition $q \in \mathbb{P}$, $q \restriction (\alpha+1)$ can be extended to a condition that decides all the values $\langle \dot{f}(\vec{\nu}): \vec{\nu} \in [g(\alpha)]^{<\omega} \rangle$, by closure). By the Fusion Lemma 2.3, there

exist $p \in G$ and a club $C \subseteq \kappa$ such that for every $\alpha \in C$ and $\vec{v} \in [g(\alpha)]^{<\omega}$, there exists a $\mathbb{P}_{\alpha+1}$ -name $\dot{\gamma}_{\vec{v}}^\alpha$ such that $p \Vdash \dot{f}(\vec{v}) = \dot{\gamma}_{\vec{v}}^\alpha$. Now, for every such $\alpha \in C$ and $\vec{v} \in [g(\alpha)]^{<\omega}$, let

$$F(\alpha, \vec{v}) = \{\gamma \in \text{Ord} : \exists r \geq p \restriction (\alpha + 1) (r \Vdash \dot{\gamma}_{\vec{v}}^\alpha = \check{\gamma})\},$$

and note that $|F(\alpha, \vec{v})| \leq |\mathbb{P}_{\alpha+1}| = \max\{\alpha^+, |h(\alpha)|\}$, due to GCH and the fact that $|\mathbb{Q}_\alpha| = |h(\alpha)|$. $\square$

Let C, F be as in Claim 5.12. Since C is a club in κ , $\kappa \in j_{E^*}(C)$. Since $a \in [j_E(g)(\kappa)]^{<\omega}$, we have that

$$\zeta = j_{E^*}(f)(a) \in (j_{E^*} \restriction V)(F)(\kappa, a).$$

Claim 5.13. *There exist a function $F' \in V$ and $\mu < j_E(g)(\kappa)$ such that*

$$\zeta = j_{E^*}(F')(\kappa, a, \mu).$$

Proof. Fix a large enough χ and a well-order $\mathcal{W}$ of H_χ such that $F \in H_\chi$.

Let us first take care of the case where $|j_{E^*}(F)(\kappa, a)| \leq \kappa^+$. In particular, there exists $\xi < \kappa^+$ such that ζ is the ξ -th element of $j_{E^*}(F)(\kappa, a)$, with respect to the $j_{E^*}(\mathcal{W})$ -least well-order of $j_{E^*}(F)(\kappa, a)$ of order-type $\leq \kappa^+$. As in Section 3, let f_ξ be the ξ -th canonical function on κ . For every $\alpha < \kappa$ and $\vec{v} \in [\alpha]^{<\omega}$, let $F'(\alpha, \vec{v})$ be the $f_\xi(\alpha)$ -th element of $F(\alpha, \vec{v})$ with respect to the $\mathcal{W}$ -least well-order of $F(\alpha, \vec{v})$ of order type $\leq \alpha^+$. It follows that $\zeta = j_{E^*}(F')(\kappa, a)$, as desired.

Thus, assume from now on that $|j_{E^*}(F)(\kappa, a)| > \kappa^+$. For every $\alpha < \kappa$ and $\vec{v} \in [\alpha]^{<\omega}$, let $\triangleleft(\alpha, \vec{v})$ be the $\mathcal{W}$ -least well-order of $F(\alpha, \vec{v})$ of minimal order type. For every $\mu < g(\alpha)$, let $F'(\alpha, \vec{v}, \mu)$ be the μ -th element in the set $F(\alpha, \vec{v})$ with respect to the well-order $\triangleleft(\alpha, \vec{v})$ (if such an element exists). Recall that

$$|j_{E^*}(F)(\kappa, a)| \leq \max\{\kappa^+, |j_{E^*}(h)(\kappa)|\},$$

by Claim 5.12. Since we assumed that $|j_{E^*}(F)(\kappa, a)| > \kappa^+$, and because $h(\alpha) \leq g(\alpha)$ for all $\alpha < \kappa$, we deduce that

$$|j_{E^*}(F)(\kappa, a)| \leq |j_{E^*}(g)(\kappa)|.$$

Since $\zeta \in j_{E^*}(F)(\kappa, a)$, there exists $\mu < j_{E^*}(g)(\kappa) \leq \lambda$ such that ζ is the μ -th element in the set $j_{E^*}(F)(\kappa, a)$, with respect to the $j_{E^*}(\mathcal{W})$ -least well-order on $j_{E^*}(F)(\kappa, a)$ of minimal order type. In particular,

$$\zeta = j_{E^*}(F')(\kappa, a, \mu).$$

Finally, since $\mu < \lambda \leq \text{crit}(k)$,

$$k(\mu) = \mu < j_{E^*}(g)(\kappa) = k(j_E(g)(\kappa)),$$

so by the elementarity of k , $\mu < j_E(g)(\kappa)$. $\square$

Let $F' \in V$ and $\mu < j_E(g)(\kappa)$ be as in Claim 5.13. It follows that

$$\zeta = (j_{E^*} \upharpoonright V)(F')(\kappa, a, \mu),$$

where κ , a , and μ are all below $j_E(g)(\kappa)$. Since we initially picked $\zeta \geq j_E(g)(\kappa)$, the above equality implies that ζ is not a generator of $j_{E^*} \upharpoonright V$, as desired.

This concludes the proof that k is the identity map.

To sum up the situation so far, we have proved that $M = M_E$ and $j_{E^*} \upharpoonright V = j_E$. Furthermore, we proved that every generator of E must lie below $j_E(g)(\kappa)$. Finally, E is κ -closed: Indeed, for every $f \in V \cap ({}^\kappa M_E)$, we have that $f \in M[H]$ (since E^* is κ -closed), namely $f \in M[H] \cap V = M_E$, where the last equality follows from Hamkins' Gap Forcing Theorem 3.7. Overall, we showed that E satisfies the hypotheses of Lemma 5.7.

Write $G = g' * G'$, where $g' \subseteq \text{Add}(\omega, 1)$ is generic over V , and $G' \subseteq \mathbb{P}$ is generic over $V[g']$. Let $H = j_{E^*}(G)$, and write $H = g' * H'$, where H' is $j_{E^*}(\mathbb{P})$ -generic over $M[g']$. Note that $j_E[G] \subseteq H$, since j_{E^*} lifts j_E .

Let $\eta = j_{E^*}(\cup G')(\kappa) < j_{E^*}(h)(\kappa)$. Again, since j_{E^*} lifts j_E , we deduce that $\eta < j_E(h)(\kappa)$. Since E satisfies the hypotheses of Lemma 5.7, the lifted extender $E_\eta^* \in V[G]$ is defined.

Let $j': V[g'] \rightarrow M[g']$ be the natural lift of j . As in the proof of Lemma 5.7, the set

$$H'_\eta = \{q \in j'(\mathbb{P}) : \exists p \in G' (q \leq j'(p) \cup \{\langle \kappa, \eta \rangle\})\}$$

is $j'(\mathbb{P})$ -generic over $M[g']$. Since $\eta = j_{E^*}(\cup G')(\kappa) = (\cup H')(\kappa)$ and $j'[G'] \subseteq H'$, we obtain that $H'_\eta \subseteq H'$, namely $H'_\eta = H'$. Thus, $\text{Ult}(V[G], E^*) \simeq M[H] = M_E[g' * H'_\eta]$ and $j_{E^*} = j_{E_\eta^*}$. It follows that E^* and E_η^* are equivalent, as desired.

This concludes the proof of Theorem 5.10. $\square$

We can now conclude the proofs of the main results of this section.

Proof of Theorem 5.4. The first clause of Theorem 5.4 follows from Lemma 5.7, and the second clause follows from Theorem 5.10. $\square$

Proof of Theorem 1.6. Fix $E \in V$ a (κ, κ^{++}) -extender witnessing that κ is $(\kappa + 2)$ -strong. For every $\alpha < \kappa$, let $g(\alpha) = \alpha^{++}$, and note that $j_E(g)(\kappa) = (\kappa^{++})^{M_E} = \kappa^{++}$, and g bounds all the generators of j_E .

Fix a cardinal $\tau \leq \kappa^{++}$. Let $h: \kappa \rightarrow \kappa$ be a function such that $h(\alpha) \leq \alpha^{++}$ for every $\alpha < \kappa$, and $j_E(h)(\kappa) = \tau$. Note that such h exists, since either $\tau \in \{\kappa^+, \kappa^{++}\}$ (in which case, h is the cardinal successor or double successor function, respectively), or $\tau < \kappa^+$ (in which case, h is the τ -th canonical function f_τ).

Let $G \subseteq \mathbb{R}^h$ be generic over V . By Theorem 5.4, E lifts in exactly $j_E(h)(\kappa) = \tau$ nonequivalent ways to an extender of $V[G]$, all of which have the same ultrapower. A standard argument (see, for instance, [10, Proposition 8.4]) shows that each of them witnesses that κ is $(\kappa + 2)$ -strong in $V[G]$. On the other hand, each (κ, κ^{++}) -extender $E^* \in V[G]$ witnessing $(\kappa + 2)$ -strongness is, again by Theorem 5.4, such a lift of some ground model (κ, κ^{++}) -extender $E \in V$. In other words, for some $\eta < j_E(h)(\kappa) = \tau$, $E^* = E_\eta^*$, $j_{E^*} \upharpoonright V = j_E$, and

$M_{E^*}^{V[G]} = M_E[g' * H_\eta]$ (where $g' \subseteq \text{Add}(\omega, 1)$ is the Cohen part of the generic G , and H_η is the $j_E(\mathbb{P})$ -generic part of $j_{E^*}(G)$). We used here the notation of Lemma 5.7). Hence

$$M_{E^*}^{V[G]} \cap V = M_E[g' * H_\eta] \cap V = M_E,$$

where the last equality follows from Hamkins' Gap Forcing Theorem 3.7. In addition, since $(V_{\kappa+2})^{V[G]} \subseteq M_{E^*}^{V[G]}$, $(V_{\kappa+2})^V \subseteq M_{E^*}^{V[G]} \cap V = M_E$. It follows that E witnesses $(\kappa + 2)$ -strongness in V . $\square$

Goldberg observed that even though there are possibly κ^{+3} -many (κ, κ^{++}) -extenders (in the GCH context), only κ^{++} of them may have the same ultrapower. In this sense, Theorem 1.6 is optimal.

Theorem 5.14 (Goldberg [16]). *Let β be the number of ultrafilters on κ . Let E be a (κ, λ) -extender. Then, up to equivalence, there are at most $\max\{\beta, \lambda\}$ extenders with the same ultrapower as E .*

Proof. Following [15], we say that an elementary embedding $i: V \rightarrow N$ is *almost an ultrapower embedding* if for every $X \subseteq N$, there exists $a \in N$ such that $X \subseteq \mathcal{H}^N(i[V] \cup \{a\})$ (here and below, $\mathcal{H}$ denotes the Skolem hull).

Claim 5.15. *Let $j: V \rightarrow M$ be an extender embedding, $i: V \rightarrow N$ be almost an ultrapower embedding, and $k: N \rightarrow M$ be an elementary embedding such that $j = k \circ i$. Then i is an ultrapower embedding, that is, there exists $a \in N$ such that $N = \mathcal{H}^N(i[V] \cup \{a\})$.¹³*

Proof. Since j is an extender embedding, there exists a set $A \in V$ such that $j(A)$ covers the set of (finite sequences of) generators of j .¹⁴ In particular, $M = \mathcal{H}^M(j[V] \cup j(A))$. It follows that $N = \mathcal{H}^N(j[V] \cup i(A))$: For every element $x \in N$, there exists a function $f \in V$ such that

$$M \models \exists a \in j(A) (k(x) = j(f)(a)),$$

where we view $k(x)$, $j(A) = k(i(A))$, and $j(f) = k(i(f))$ as parameters. By the elementarity of k ,

$$N \models \exists a \in i(A) (x = i(f)(a)).$$

So $x \in \mathcal{H}^N(j[V] \cup i(A))$.

Since i is almost an ultrapower embedding, for every ordinal ξ , there exists $b_\xi \in N$ such that $(V_\xi)^N \subseteq \mathcal{H}^N(j[V] \cup \{b_\xi\})$. Each such b_ξ belongs to $\mathcal{H}^N(j[V] \cup \{a_\xi\})$ for some $a_\xi \in i(A)$. Thus, there exists $a \in i(A)$ such that for unboundedly many ordinals ξ , $b_\xi \in \mathcal{H}^N(j[V] \cup \{a\})$. Thus, $N \subseteq \mathcal{H}^N(j[V] \cup \{a\})$. $\square$

¹³We refer the interested reader to [15, Section 2.1] for further details on Goldberg's characterization of ultrapower and extender embeddings in terms of Skolem hulls.

¹⁴Since an extender embedding cannot have a proper class of generators, we can take $A = V_\chi$ for a large enough χ .

Keeping this in mind, fix an extender E as in the formulation of Theorem 5.14. By [15, Theorem 3.7], for every extender F with the same ultrapower as E , there exists an inner model N (actually, $N = \mathcal{H}^M(j_E[V] \cup j_F[V])$), elementary embeddings $i_0: V \rightarrow N$, $i_1: V \rightarrow N$, and $k: N \rightarrow M_E$, such that $j_E = k \circ i_0$, $j_F = k \circ i_1$, and i_0, i_1 are almost ultrapower embeddings. By the above claim, since i_0, i_1 are factors of extender embeddings, both are ultrapower embeddings with respect to ultrafilters Z_0, Z_1 on κ , respectively.

We argue that k is uniquely determined by E , a function $f: \kappa \rightarrow \kappa$, and some $a \in [\lambda]^{<\omega}$. Indeed, let $a \in [\lambda]^{<\omega}$ and $f \in V$ be such that $k([id]_{Z_0}) = j_E(f)(a)$. Then $Z_0 = f_*(E_a) = \{X \subseteq \kappa: f^{-1}[X] \in E_a\}$, and k maps each element of the form $i_0(g)([id]_{Z_0})$ to $j_E(g)(j_E(f)(a))$. Thus, k is uniquely determined by E , f , and a .

Since $j_F = k \circ j_{Z_1}$, F is uniquely determined by Z_1 , E , f , and $a \in [\lambda]^{<\omega}$. The set of all such parameters has size $\max\{\beta, \lambda\}$. $\square$

We finish this section with a complete classification of normal measures in $V[G]$, for $G \subseteq \mathbb{R}$ generic over V . By Remark 5.9, normal measures $W \in V[G]$ may arise from non-normal measures $E \in V$ which have an additional single generator above κ . In this case, the forcing $\mathbb{P}$ might “kill” this additional generator, and E lifts to a normal measure which differs from those described in Theorem 1.2.¹⁵

We summarize this in the following theorem.

Theorem 5.16. *Assume GCH. Let $G \subseteq \mathbb{R}$ be generic over V . Suppose that $W \in V[G]$ is a normal measure on κ . Then there exists a $(\kappa, j_W(g)(\kappa))$ -extender $E \in V$ such that $W = E_\eta^*$ for some $\eta < j_E(h)(\kappa)$. Furthermore, E is equivalent to a κ -complete ultrafilter on κ , which is the one derived from j_E using (κ, η) as a seed.*

Proof. As before, write $G = g' * G'$, where $g' \subseteq \text{Add}(\omega, 1)$ is generic over V , and $G' \subseteq \mathbb{P}$ is generic over $V[g']$.

Fix a well-order $\mathcal{W}$ of V_κ . Then $\mathcal{W}$ induces a well-order $\mathcal{W}'$ of $(V_\kappa)^{V[g']}$ (by ordering elements with respect to their $\mathcal{W}$ -least $\text{Add}(\omega, 1)$ -names).

Let $\lambda = j_W(g)(\kappa) < j_W(\kappa)$. Let E^* be the (κ, λ) -extender derived from j_W . Note that $j_W = j_{E^*}$, since the only generator of W is κ , and it is below $j_W(g)(\kappa)$. By Theorem 5.10, we deduce that $j_W = j_{E_\eta^*}$, for some κ -closed (κ, λ) -extender $E \in V$ and $\eta < j_E(h)(\kappa)$. By the analysis in the proof of Theorem 5.10, $\eta = j_W(\cup G')(\kappa)$.

It remains to prove that every $x \in M_E = \text{Ult}(V, E)$ has the form $j_E(\kappa, \eta)$, and thus E is equivalent to the measure derived from j_E using (κ, η) as a seed.¹⁶

¹⁵Indeed, every normal measure in the generic extension, in the context of Theorem 1.2, yields an ultrapower embedding that lifts an ultrapower embedding arising from a normal measure in V . In contrast, in Theorem 5.16, a normal measure in the generic extension may induce an elementary embedding whose restriction to V is an ultrapower embedding derived from a non-normal measure.

¹⁶Possibly, E is already equivalent to a normal measure, and then κ is the single generator of j_E and W is really a lift of a ground model normal measure.

Fix $\zeta \neq \kappa, \eta$. Since W is normal, there exists $f \in V[G]$ such that $\zeta = j_W(f)(\kappa)$.

Work in $V[g']$. For every $\alpha < \kappa$, let $e(\alpha) \subseteq \mathbb{P}$ be the dense open set of conditions deciding the value of $\dot{f}(\alpha)$. By applying the Fusion Lemma 2.3 on the sequence $\langle e(\alpha) : \alpha < \kappa \rangle$, there exist $p' \in G'$ and a club $C \subseteq \kappa$ in $V[g']$ such that for every $\alpha \in C$,

$$\{r \in \mathbb{P}_{\alpha+1} : \exists \gamma \in \text{Ord} \left(r \restriction (p' \setminus (\alpha+1)) \Vdash \dot{f}(\alpha) = \check{\gamma} \right) \}$$

is a dense open subset of $\mathbb{P}_{\alpha+1}$ above $p' \restriction (\alpha+1)$. In other words, for every $\alpha \in C$,

$$p' \restriction (\alpha+1) \Vdash \exists \gamma \in \text{Ord} \left(p' \restriction (\alpha+1) \Vdash \dot{f}(\alpha) = \check{\gamma} \right).$$

Let $j' : V[g'] \rightarrow M_E[g']$ be the natural lift of j_E . Since $\kappa \in j'(C)$, we have in $M_E[g']$ that

$$p' \restriction 0_{\mathbb{Q}_\kappa} \Vdash \exists \gamma \in \text{Ord} \left(j'(p') \restriction (\kappa+1) \Vdash j'(\dot{f})(\kappa) = \check{\gamma} \right),$$

where $\mathbb{Q}_\kappa = j'(\langle \mathbb{Q}_\alpha : \alpha < \kappa \rangle)(\kappa)$. Next, since $\eta < j_E(h)(\kappa)$ induces a $\mathbb{Q}_\kappa$ -generic set over $M_E[g'][G']$, we deduce that

$$p' \Vdash \exists \gamma \in \text{Ord} \left(\{ \langle \kappa, \eta \rangle \} \cup j'(p') \restriction (\kappa+1) \Vdash j'(\dot{f})(\kappa) = \check{\gamma} \right).$$

Since E is κ -closed, ${}^\kappa M_E[g'] \subseteq M_E[g']$, and thus $j'(\mathbb{P})_\kappa = \mathbb{P}$. Therefore, in $M_E[g']$, there exists a $\mathbb{P}$ -name $\dot{\gamma}$ such that

$$j'(p') \cup \{ \langle \kappa, \eta \rangle \} \Vdash j'(\dot{f})(\kappa) = \dot{\gamma}.$$

We can assume that $\dot{\gamma}$ is the $j'(\mathcal{W})$ -least $\mathbb{P}$ -name satisfying the formula in the previous display. It follows that $\dot{\gamma}$ has the form $j'(\ell)(\kappa, \eta)$ for some function $\ell \in V[g']$: Indeed, over $V[g']$, let $\ell(x, y)$ be the $\mathcal{W}$ -least $\mathbb{P}_x$ -name $\dot{\gamma}$ such that $p' \cup \{ \langle x, y \rangle \} \Vdash \dot{f}(\alpha) = \dot{\gamma}$.

Next, work over $M_E[g']$. Since $\dot{\gamma} \in M_E[g']$ is a $\mathbb{P}$ -name for an ordinal, we can define the set A of all possible values of $\dot{\gamma}$, namely,

$$A = \{ \beta \in \text{Ord} : \exists q \geq p' \ (q \Vdash \dot{\gamma} = \check{\beta}) \}.$$

Then $A \in M_E[g']$ has the form $j'(\ell^*)(\kappa, \eta)$ for some function $\ell^* \in V[g']$: Indeed, let $\ell^*(x, y)$ be the set $\{ \beta \in \text{Ord} : \exists q \geq p' \restriction x \ (q \Vdash \ell(\check{x}, \check{y}) = \check{\beta}) \}$. Note that $|A| \leq \kappa^+$, since by GCH, $|\mathbb{P}| = \kappa^+$. Also, the definitions above ensure that over $M_E[g']$,

$$j'(p') \cup \{ \langle \kappa, \eta \rangle \} \Vdash j'(\dot{f})(\kappa) \in \check{A}.$$

Since $p' \in G'$ and $\eta = j_W(\cup G')(\kappa)$, we deduce that in $\text{Ult}(V[G], W) \simeq M_E[g'][j_W(G')]$,

$$\zeta = j_W(f)(\kappa) \in A.$$

In particular, for some ordinal $\xi < \kappa^+$, ζ is the ξ -th element of A with respect to the $j'(\mathcal{W})$ -least well-order of A of order type $\leq \kappa^+$. As before, letting f_ξ be the canonical function that represents ξ , we obtain that, in $M_E[g']$, ζ is the $j'(f_\xi)(\kappa)$ -th element of $j'(\ell^*)(\kappa, \eta)$, with respect to the $j'(\mathcal{W})$ -least well-order of $j'(\ell^*)(\kappa, \eta)$ of order type $\leq \kappa^+$.

All of this shows that $\zeta = j'(F')(\kappa, \eta)$ for some function $F' \in V[g']$. In particular, there exists a function $F \in V$ such that $\zeta = j_E(F)(\kappa, \eta)$: Indeed, let $\dot{F}'$ be an $\text{Add}(\omega, 1)$ -name for F' , and let $s \in g'$ be a condition such that, in M_E , $s \Vdash \dot{\zeta} = j_E(\dot{F}')(\check{\kappa}, \check{\eta})$. For every x, y , let $F(x, y)$ be the unique ordinal δ such that $s \Vdash \dot{\delta} = \dot{F}'(\check{x}, \check{y})$, if such an ordinal exists.

Since ζ was an arbitrary ordinal outside of $\{\kappa, \eta\}$, we deduce that E is equivalent to the κ -complete ultrafilter derived from j_E using (κ, η) as a seed. $\square$

Remark 5.17. The case where $h(\alpha) \leq \alpha^+$ for every $\alpha < \kappa$ is similar to the one exhibited in Theorem 1.2. This is due to the fact that every $\eta < j_E(h)(\kappa)$ satisfies $\eta < \kappa^+$, and in particular, η has the form $j_E(f_\eta)(\kappa)$ for the canonical function f_η . In other words, the measure derived from j_E using (κ, η) as a seed is equivalent to a normal measure. Thus, by Theorem 5.16, every normal measure $W \in V[G]$ has the form U_η^* for a normal measure $U \in V$ and $\eta < j_U(h)(\kappa)$.

6 Open questions and concluding remarks

In this final section, we discuss some open questions.

Question 6.1. Is it consistent that GCH holds, there are κ^{++} -many normal measures on κ , but only a single normal ultrapower?

The same proof technique of Theorem 1.2 does not appear to apply here (see Remark 3.11), but we still conjecture that the answer is positive.

Note that a positive answer to Question 6.1 would exhibit behavior somewhat different from that of extenders. For example, assuming GCH, there are at most κ^{++} -many (κ, κ^{++}) -extenders with the same ultrapower (by Theorem 5.14), even though the total number of (κ, κ^{++}) -extenders can be κ^{+3} .

Next, recall that the Friedman–Magidor theorem generalizes well to the context of the violation of GCH at a measurable cardinal.

Theorem 6.2 (Friedman and Magidor, [11]). *Let $V = L[\bar{E}]$ be the minimal extender model that contains a cardinal κ which is $(\kappa + 2)$ -strong. Then for every cardinal $\tau \leq \kappa^{++}$, there exists a cofinality preserving forcing extension in which κ is measurable, $2^\kappa = \kappa^{++}$, and there are exactly τ normal measures on κ .*

This motivates the question whether the methods developed in this paper can yield similar results.

Question 6.3. Assume that κ is $(\kappa + 2)$ -strong in a model $V \models \text{ZFC} + \text{GCH}$. Fix a cardinal $\tau \leq \kappa^{++}$. Is there a cardinal preserving forcing extension $V[G]$ such that the following hold?

1. κ is measurable and $2^\kappa = \kappa^{++}$ in $V[G]$.
2. For every extender $E \in V$ witnessing that κ is $(\kappa + 2)$ -strong, there are τ -many normal measures on κ in $V[G]$ such that for each of them, say U , $j_U^{V[G]}$ lifts j_E .
3. Every normal measure on κ in $V[G]$ arises as a lift of some $(\kappa + 2)$ -strong extender E .

In [7], it was shown that the Ultrapower Axiom is consistent with the failure of GCH at a measurable cardinal, via the construction of an $L[U]$ -like model, i.e., a model in which κ is the unique measurable cardinal, κ carries a single normal measure U , and every other σ -complete ultrafilter is equivalent to U^n for some $n < \omega$. This result motivates the following question:

Question 6.4. Is the Weak Ultrapower Axiom and the negation of the Ultrapower Axiom consistent with the failure of GCH at a measurable cardinal?

A natural approach to Question 6.4 is to force with the splitting forcing over the $L[U]$ -like model described above and analyze all possible lifts of U^n , as in Section 4. Indeed, the proofs of Theorems 1.4 and 1.5 confirm that this construction yields a model of the Weak UA. However, it is unclear whether κ^{++} remains a cardinal at the end of the process. This is due to the fact that applying the splitting forcing to a ground model in which GCH already fails at κ may collapse κ^{++} and inadvertently restore GCH, as the following claim shows.

Claim 6.5. *Assume that in V κ is Mahlo, $2^\kappa = \kappa^{++}$, and $2 \leq \tau \leq \kappa^+$ is a cardinal. Let $\mathbb{P} = \mathbb{P}^\tau$ be the splitting forcing. Suppose there exists a collection $\mathcal{C} \in V$ of clubs in κ such that $|\mathcal{C}| = \kappa^+$, and every club $D \subseteq \kappa$ in V contains a club from $\mathcal{C}$.¹⁷ Then the splitting forcing $\mathbb{P}$ collapses κ^{++} .¹⁸*

Proof. Let $G \subseteq \mathbb{P}$ be generic over V . We claim that in $V[G]$, there exists a surjection from $\mathcal{C}$ to $(\mathcal{P}(\kappa))^V$.

We say that for $C \in \mathcal{C}$ and $A \subseteq \kappa$, the generic filter G codes A via C if, for every $i < \kappa$,

$$i \in A \iff (\cup G)(C_i^+) = 1,$$

where C_i^+ denotes the i -th element in the set of inaccessible cardinals in C which are not limits of inaccessible cardinals in C . Since κ is Mahlo, C_i^+ exists for every $i < \kappa$.

¹⁷These assumptions are consistent. See, for example, [12].

¹⁸A similar argument shows that any other reasonable nonstationary support product or iteration collapses κ^{++} under the same assumptions.

Every club $C \in \mathcal{C}$ codes via G a unique $A \subseteq \kappa$, but it's not necessarily true that $A \in V$. Define in $V[G]$ the partial function $\varphi: \mathcal{C} \rightarrow (\mathcal{P}(\kappa))^V$, which maps each $C \in \mathcal{C}$ to the set $A \in (\mathcal{P}(\kappa))^V$ such that G codes A via C , if such a set A exists.

To conclude the proof of Claim 6.5, it suffices to show that for every $A \subseteq \kappa$ in V , there exists $C \in \mathcal{C}$ such that G codes A via C . Hence, φ is a surjection from a set having size $(\kappa^+)^V$ to a set having size $(\kappa^{++})^V$. It therefore follows that $\mathbb{P}$ collapses $(\kappa^{++})^V$.

Indeed, fix $A \subseteq \kappa$ in V and a condition $p \in \mathbb{P}$. By assumption, there is a club $C \in \mathcal{C}$ disjoint from $\text{supp}(p)$. The sequence $\langle C_i^+ : i < \kappa \rangle$ is nowhere stationary,¹⁹ so we may extend p to a condition $q \in \mathbb{P}$ by defining

$$q = p \cup \{ \langle C_i^+, 1 \rangle : i \in A \} \cup \{ \langle C_i^+, 0 \rangle : i \notin A \}.$$

By density, there exists $q \in G$ forcing that G codes A via some $C \in \mathcal{C}$.

This shows that φ is a surjection, thereby proving Claim 6.5. $\square$

Claim 6.5 naturally leads to the following question:

Question 6.6. Can the splitting forcing preserve κ^{++} over a ground model V in which $2^\kappa = \kappa^{++}$? Alternatively, is there a variant of the splitting forcing that preserves κ^{++} , even when $2^\kappa = \kappa^{++}$ in V ?

Finally, the following central question, raised by Friedman and Magidor in [11], has remained open ever since:

Question 6.7. Is there a Friedman–Magidor theorem for fine, normal measures on $\mathcal{P}_\kappa(\lambda)$ for $\lambda > \kappa$?

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¹⁹This holds, since the function f that maps each C_i^+ to the supremum of the inaccessible cardinals in C below it (and C_0^+ is mapped to 0) is regressive and injective. Hence, if there were an inaccessible cardinal $\alpha \leq \kappa$ such that $\{C_i^+ : i < \kappa\} \cap \alpha$ is stationary in α , by Fodor's lemma, there would be $i_0 \neq i_1$ such that $f(C_{i_0}^+) = f(C_{i_1}^+)$, contradicting the fact that f is injective on $\{C_i^+ : i < \kappa\}$.

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